

An Econometric Assessment of Visa Inc.'s Share Price Using Multivariable Regression and Time-Series Modelling

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Abstract

Visa Inc.'s equity performance offers a comprehensive view through which broader macro-financial developments can be observed, given the firm's reliance on global consumption activity, employment conditions, household income growth, payment flows, and overall financial stability. Using a monthly dataset that integrates market, economic, and sentiment-based indicators, the analysis explores how shifts in the macroeconomy translate into movements in Visa's share price. The results highlight clear associations between Visa's valuation and a range of economic forces linked to consumer behaviour, the strength of the labour market, monetary policy conditions, financial stress, and changes in household purchasing power. Although the larger group of variables contains useful economic information, the diagnostic checks reveal that some of them overlap, some behave inconsistently, and the residuals exhibit patterns that render the full model less reliable. Refining the set of factors gives a more stable and reliable picture of what truly drives Visa's monthly price movements. It keeps the important economic relationships while removing the extra variables that add little useful information.

Alongside the longer-term economic factors, the study also finds that Visa's share price shows patterns seen in financial markets, which include periods where prices move more persistently, times when they react strongly to changes in risk conditions, and moments of increased volatility. Looking at returns shows that these movements often line up with changes in investor mood, expectations about future policy, and shifts in the wider economic cycle. This highlights the need to consider both broad economic influences and shorter-term market behaviour when trying to understand how a company in the payments sector is valued. The results show that Visa's share price is affected by several different forces at the same time, including underlying economic conditions and the changing environment of financial markets.

Academic Honesty Statement

"I have neither given nor received help (apart from the instructor) to complete this assignment."

Signature:

Deepak Chaudhary

I. Background information (VISA Inc)

Visa Inc. is a global payments technology company that operates one of the world's most extensive electronic payments networks. Its core platform, VisaNet, enables authorisation, clearing, and settlement for card-based transactions across more than 200 countries and territories, supporting billions of payment interactions each year (Visa Inc., 2024a). The network is designed to provide high levels of speed, security, and reliability, allowing consumers, merchants, financial institutions, and governments to undertake digital payments with minimal friction.

A defining feature of Visa's business model is that it does not act as a bank. The company does not issue cards, extend credit, or hold deposits; these responsibilities are handled by issuing financial institutions. Instead, Visa earns revenue primarily through service revenues generated from payment volumes, and data-processing revenues arising from the number of transactions it processes (Visa Inc., 2024b). This fee-based structure enables Visa to maintain a scalable operating model driven by the level of global commerce and electronic payment usage rather than balance-sheet risk.

Visa's financial performance, therefore, has a close connection to broader macroeconomic and financial conditions. Growth in household spending, employment, income, and consumer confidence generally leads to increased payment activity, while periods of economic weakness, uncertainty, or tighter financial conditions can reduce transaction flows and slow revenue growth. The company's cross-border business, which includes transactions where the merchant and the card issuer are located in different countries, is particularly influenced by international travel patterns, exchange-rate movements, and global economic conditions (Visa Inc., 2024b). As these transactions tend to carry higher fees, changes in global mobility and financial conditions can affect Visa's profitability and, in turn, investor expectations regarding the company's valuation.

Given its global footprint, reliance on payment volumes, and sensitivity to both domestic and international economic activity, Visa provides a suitable basis for examining how macroeconomic and financial developments relate to movements in a large payments-sector equity. Its business model offers a relatively clear link between economic indicators, including labour-market conditions, consumer spending, inflation, and financial-market sentiment and the company's operating performance, making it a relevant case for understanding the transmission of macro-financial information into equity prices.

II. Description of the Factors:

This study is based on a monthly dataset covering the period from January 2020 to June 2025. The information is drawn from two main sources: Yahoo Finance for market-price data and the Federal Reserve Economic Data (FRED) database for macroeconomic and financial indicators. The dataset brings together twenty-four variables that capture movements in financial markets, household spending patterns, labour-market conditions, broader economic activity, and measures of financial stress. All series are aligned at a monthly frequency to remain consistent with Visa's adjusted closing price.

Price data for Visa's shares, along with major market indices such as the S&P 500 (GSPC) and NASDAQ Composite (IXIC), were taken from Yahoo Finance, as were observations for the VIX, the US Dollar Index (DXY), gold prices (GLD), and Bitcoin (BTC-USD). The wider set of economic indicators, including retail sales, personal consumption expenditure, industrial production, unemployment, personal income, consumer sentiment, outstanding consumer credit, inflation measures, Treasury yields, the Chicago Fed Financial Conditions Index, payroll numbers, real disposable income, and housing-market activity, was collected from the FRED database.

Together, these variables give a rounded picture of the economic environment in which Visa operates. They cover developments in consumer demand, financial-market behaviour, employment, income, price levels, and interest-rate conditions. Bringing these indicators into a single dataset allows for a clearer assessment of how changes in the macro-financial landscape are associated with movements in Visa's monthly share price.

Data Sources:

Equity and market data. Monthly data for Visa's share price and trading volume were obtained from Yahoo Finance, a well-known platform that provides stock price data. Using the quantmod package, we downloaded the adjusted closing price and trading volume for Visa (ticker V), together with closing prices for the S&P 500 index (^GSPC), NASDAQ Composite index (^IXIC), CBOE Volatility Index (^VIX), U.S. Dollar Index (DX-Y.NYB), gold price (GLD), and Bitcoin (BTC-USD). Log differences of these series were used to construct monthly returns.

Macroeconomic and financial indicators. A broad set of monthly macro-financial variables was retrieved from the Federal Reserve Bank of St Louis's FRED database, fredhelp.stlouisfed.org. FRED is an online repository that curates hundreds of thousands of economic time series from national and international sources. The series used in this analysis includes:

- Real economy indicators: retail sales (RSAFS), personal consumption expenditure (PCE), industrial production (INDPRO), payroll employment (PAYEMS), and housing starts (HOUST).
- Labour and income variables: unemployment rate (UNRATE), personal income (PI), real disposable income (DSPIC96) and consumer sentiment (UMCSENT).
- Credit and financial conditions: total consumer loans (TOTCI) and the Chicago Fed National Financial Conditions Index (NFCI).
- Inflation and interest rates: consumer price index (CPIAUCSL), its month-on-month change, the 10-year Treasury yield (DGS10) and the 2-year Treasury yield (DGS2).
- Commodity prices: West Texas Intermediate crude oil price (DCOILWTICO), gold price (via GLD), and Bitcoin price (CBBTCUSD).

All series were converted to end-of-month frequency using monthly() and merged into a single dataset covering December 2019 to June 2025. This combined panel provides a balanced view of financial markets and macroeconomic conditions that influence Visa's valuation.

Dependent Variable:

1. **VISA_Price:** This is the monthly adjusted closing price of Visa Inc. It reflects how the market values the company in light of consumer spending trends, payment-volume growth, financial-market conditions, and expectations about future economic activity. As Visa earns revenue from transaction fees, its share price responds strongly to changes in spending levels and key macroeconomic indicators.

Independent Variables:

2. **VISA_Volume:** This measures the number of Visa shares traded each month. Higher trading volume usually reflects increased investor attention or reactions to financial news. Sudden increases in volume often occur during earnings announcements, economic shocks, or periods of market volatility, which can amplify short-term movements in Visa's share price.
3. **SP500_Return:** The S&P 500 return reflects the performance of the broad United States equity market. When the index rises, it generally indicates stronger economic conditions and healthier

corporate earnings, which tend to support higher consumer spending. Visa benefits from increased payment activity in such periods, making this variable positively related to its share price.

4. **Nasdaq_Return:** The NASDAQ index tracks a large group of technology-oriented companies. A strong performance in this index often reflects growth in digital services, e-commerce, and the adoption of financial technology. As Visa is a leading provider of digital payment infrastructure, developments in the technology sector can support higher transaction volumes and may contribute to an increase in Visa's valuation.
5. **VIX_Level:** The VIX measures expected market volatility and uncertainty. When it increases, households and businesses often reduce discretionary spending, which can lead to fewer card transactions. High VIX readings typically place downward pressure on Visa's share price because they reflect weaker confidence and tighter financial conditions.
6. **DXY_Return:** The United States Dollar Index reflects the strength of the dollar relative to a basket of major currencies. Visa generates a significant portion of its revenue from cross-border transactions. When the dollar becomes particularly strong, international spending may decline, reducing high-margin cross-border payment volumes and putting pressure on Visa's share price.
7. **Gold_Return:** Gold prices tend to rise during periods of uncertainty or increased risk aversion. Higher gold returns often signal worsening market sentiment. During such times, consumers usually cut back on non-essential spending, which can reduce card-transaction volumes. Weaker sentiment of this kind is often associated with lower revenues for Visa and a decline in its share price.
8. **Bitcoin_Return:** Bitcoin returns capture shifts in speculative risk appetite within financial markets. Sharp movements in cryptocurrency prices can influence spending preferences, interest in digital assets, and general market sentiment. Although the relationship with Visa is indirect, large swings in Bitcoin returns may affect investor behaviour towards digital payment firms and technology-related companies, including Visa.
9. **Oil_Price:** Higher oil prices raise essential living costs, particularly fuel expenditure. When households devote a greater share of income to necessities, discretionary spending tends to fall. This can lead to lower transaction volumes on Visa cards, which may place downward pressure on Visa's revenue growth and share price performance.
10. **Retail_Sales:** Retail sales provide a direct indication of monthly consumer spending. Increases in retail activity typically lead to a higher number of card transactions processed across Visa's network. Strong retail sales, therefore, tend to support Visa's share price, as they signal robust payment-volume growth.
11. **PCE_Consumption:** Personal Consumption Expenditure (PCE) is one of the most comprehensive measures of household spending in the United States. As Visa earns revenue from processing consumer transactions, higher PCE values generally support stronger financial performance and tend to increase the company's share valuation.

- 12. Industrial_Production:** Industrial production tracks output across manufacturing and related industrial sectors. Strong performance in this measure usually indicates a healthier economy, leading to more employment, higher income, and increased consumer spending. These developments indirectly support Visa's revenue through greater payment activity.
- 13. Unemp_Rate:** A higher unemployment rate reduces household income and limits spending capacity. When joblessness increases, consumers often scale back their use of credit cards and discretionary purchases. This lowers the number of transactions processed on Visa's network, which can negatively affect its share price.
- 14. Payroll_Employment:** Payroll employment measures the number of people in paid employment. Higher employment levels typically imply increased income, greater financial stability, and stronger consumer spending. As employment rises, Visa benefits from higher transaction volumes, which usually support its share performance.
- 15. Personal_Income:** Higher personal income increases household purchasing power. When people earn more, they typically spend more, leading to a greater number of Visa card transactions. Strong income growth is therefore generally supportive of Visa's share performance.
- 16. Real_Disposable_Income:** Real disposable income measures the amount of money households have available to spend after taxes, adjusted for inflation. It reflects true purchasing power. When this measure rises, Visa often benefits from higher transaction volumes across retail, services, and travel-related categories.
- 17. Consumer_Sentiment:** Consumer sentiment reflects how confident households feel about current and future economic conditions. Higher sentiment tends to encourage spending and card usage. When confidence strengthens, Visa's revenue prospects improve, which can support its share valuation.
- 18. Consumer_Loans:** An increase in consumer borrowing raises spending capacity. Greater use of credit typically results in more transactions, higher card volumes, and increased fee revenue for Visa. Growth in consumer loans is therefore usually positive for the company's share performance.
- 19. CPI_Index:** The Consumer Price Index measures overall inflation. Higher inflation reduces real purchasing power, limits discretionary spending, and may dampen transaction activity. Prolonged periods of inflation can put downward pressure on Visa's share price.
- 20. CPI_Mom:** The month-on-month Consumer Price Index captures short-term changes in inflation. Sudden increases reduce real income in the near term, making households more cautious with their spending. High CPI-MoM figures can temporarily reduce transaction volumes and weigh on Visa's performance.

- 21. Y10Y (10-Year Treasury Yield):** The ten-year Treasury yield is a widely used benchmark for long-term interest rates. Increases in this yield can tighten financial conditions and raise borrowing costs. When long-term rates rise sharply, economic activity may soften, leading to reduced consumer spending and slower growth in Visa's transaction volumes.
- 22. Y2Y (2-Year Treasury Yield):** The two-year Treasury yield reflects expectations of short-term monetary policy. A rise in this yield often signals forthcoming policy tightening. Such conditions can restrict credit availability, dampen household spending, and place downward pressure on Visa's share price.
- 23. Financial_Conditions_Index (NFCI):** The National Financial Conditions Index measures levels of stress across credit, liquidity, and funding markets. When financial conditions tighten, both consumers and businesses tend to reduce spending, and borrowing becomes more costly. These effects usually lead to lower transaction volumes on Visa's network and may reduce the company's revenue and share performance.
- 24. Housing_Starts:** Housing starts measure new residential construction and serve as an indicator of economic confidence and household formation. When construction activity increases, spending typically rises on items such as furniture, building materials, home appliances, and related services. These purchases generate a large number of card transactions, which can positively influence Visa's overall performance.

III. Structure of the Multivariable Linear Regression Model

The purpose of the full regression model is to examine how a broad range of economic, financial, and market indicators relate to the monthly share price of Visa Inc. As Visa's performance is closely tied to consumer spending, income conditions, financial stress, and movements in interest rates, the model incorporates all variables that could reasonably influence payment activity or shape investor expectations. Beginning with a wide specification helps ensure that no potentially important factor is omitted. These variables jointly capture changes in consumer behaviour, labour-market strength, credit availability, inflation pressure, spending power, and financial conditions, all of which affect Visa's transaction volumes and overall valuation.

The full regression model is expressed as:

$$\begin{aligned} \text{VISA_Price}_t = & \beta_0 + \beta_1 \text{VISA_Volume}_t + \beta_2 \text{SP500_Return}_t + \beta_3 \text{Nasdaq_Return}_t \\ & + \beta_4 \text{VIX_Level}_t + \beta_5 \text{DXY_Return}_t + \beta_6 \text{Gold_Return}_t + \beta_7 \text{Bitcoin_Return}_t \\ & + \beta_8 \text{Oil_Price}_t + \beta_9 \text{Retail_Sales}_t + \beta_{10} \text{PCE_Consumption}_t \\ & + \beta_{11} \text{Industrial_Production}_t + \beta_{12} \text{Unemp_Rate}_t + \beta_{13} \text{Personal_Income}_t \\ & + \beta_{14} \text{Consumer_Sentiment}_t + \beta_{15} \text{Consumer_Loans}_t + \beta_{16} \text{CPI_Index}_t \\ & + \beta_{17} \text{CPI_Mom}_t + \beta_{18} \text{Y10Y}_t + \beta_{19} \text{Y2Y}_t + \beta_{20} \text{Financial_Conditions_Index}_t \\ & + \beta_{21} \text{Payroll_Employment}_t + \beta_{22} \text{Real_Disposable_Income}_t \\ & + \beta_{23} \text{Housing_Starts}_t + \varepsilon_t \end{aligned}$$

where

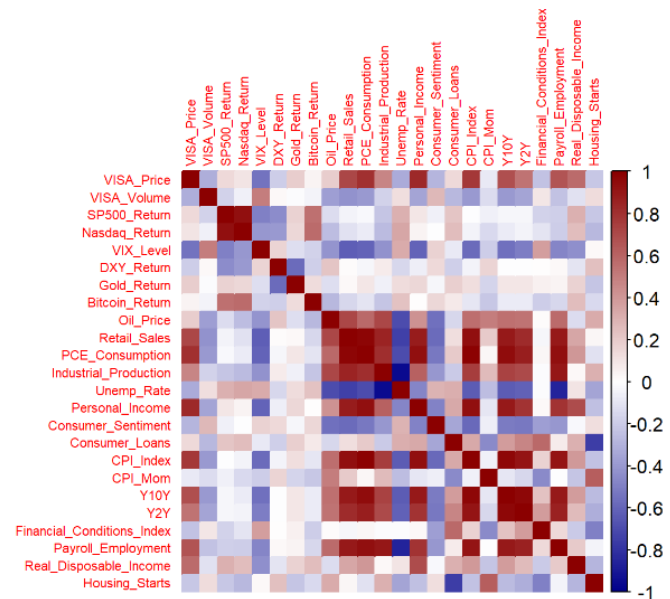
β_0 is the intercept, $\beta_1, \dots, \beta_{23}$ measure the effect of each predictor, and ε_t is the error term.

IV. Diagnostic Testing and Gauss-Markov Assumptions

Several diagnostic checks are carried out to assess whether the full regression model satisfies the Gauss-Markov assumptions. Linearity is examined using the residual-versus-fitted plot, and the normality of the residuals is assessed through the Q-Q plot. Homoscedasticity is tested using both the Breusch-Pagan test and White's test, while the Durbin-Watson statistic is used to evaluate independence of the error terms. Multicollinearity is measured through Variance Inflation Factors, which provide an indication of how strongly each predictor is related to the others.

Figure A shows a correlation heat map illustrating the pairwise relationships among all independent variables. The chart reveals numerous strong correlations between key macroeconomic indicators, including labour-market measures, income variables, interest-rate series, and consumer-activity metrics. These pronounced correlations help to account for the very high VIF values in the full model and demonstrate that multicollinearity is a significant concern. Although the broad specification captures meaningful variation in Visa's monthly share price, the diagnostic results point to issues such as residual autocorrelation, heteroscedasticity, and substantial overlap among predictors. These findings support the need for a reduced model that provides a more statistically stable and interpretable representation of the factors associated with Visa's price movements.

Figure A: Correlation Heatmap of VISA Predictors



V. Hypothesis Testing of the Regression Coefficients:

A. Tests of Individual Coefficients (t-tests):

Under the Gauss-Markov assumptions, each coefficient β_i in the complete model can be tested for individual statistical significance using the standard t-test. For any predictor X_i , the hypotheses are:

$$H_0: \hat{\beta}_i = 0$$

$$H_1: \hat{\beta}_i \neq 0$$

The test statistic is:

$$T_{\text{STAT}} = \frac{\hat{\beta}_i}{SE(\hat{\beta}_i)}$$

where $SE(\hat{\beta}_i)$ is the standard error of the estimated coefficient.

Decision Rule:

We reject H_0 if $|T_{\text{STAT}}| > t_{\text{critical}, \alpha/2}$.

Taking an example of testing the significance of Unemployment Rate (Unemp_Rate):

$$\hat{\beta}_{\text{Unemp}} = 23.20, SE = 5.755$$

$$T_{\text{STAT}} = \frac{23.20}{5.755} = 4.031$$

We reject H_0 if $|T_{\text{STAT}}| > t_{42, 0.025}$.

For the full model, degrees of freedom are:

$$n - k - 1 = 66 - 23 - 1 = 42.$$

The critical value is $t_{42,0.025} = 2.02$.

Since:

$$|4.031| > 2.02,$$

we reject H_0 and conclude that the Unemployment Rate is statistically significant at the 5% level.

A look at the individual t-statistics reveals that only a few variables appear significant at the 5% level in the full model. Even so, it is not appropriate to rely solely on these significance levels because the full specification is heavily affected by multicollinearity. This inflates standard errors and makes several variables seem unimportant when their effects are simply obscured (Wooldridge, 2016). For this reason, the reduced VISA model is developed by combining statistical evidence with economic understanding, rather than depending purely on the p-value. The chosen variables include PCE Consumption, Unemployment Rate, the 2-Year Treasury Yield, the Financial Conditions Index, and Payroll Employment. These predictors either remain statistically significant once highly correlated variables are removed or represent important economic and financial measures closely linked to Visa's performance. In contrast, the other variables show unstable coefficients, higher p-values, and limited explanatory contribution, mainly because of the multicollinearity in the full model. This shows that many of the insignificant t-statistics arise from multicollinearity rather than a lack of economic importance, even though the F-statistic indicates strong joint significance.

B. Joint Significance of Coefficients (F-test):

In this section, we test whether all twenty-three independent variables in the full Visa regression model are jointly statistically significant in explaining the dependent variable, VISA_Price. Under the Gauss-Markov assumptions, the appropriate method is the F-test for overall significance (Gujarati and Porter, 2009).

We test whether the predictors, taken together, have explanatory power:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \dots = \beta_{23} = 0$$

$$H_1: \text{At least one } \beta_i \neq 0, i = 1, 2, \dots, 23$$

Under the null hypothesis, none of the variables contributes to explaining changes in Visa's stock price.

From the R output of the full model, the F-statistic is:

$$F_{\text{STAT}} = 34.62$$

With degrees of freedom:

Numerator df (k) = 23; Denominator df (n-k-1) = 66 - 23 - 1 = 42 at the 5% significance level

The critical value from the F-distribution is:

$$F_{\text{CRITICAL}} = F_{(23,42), 0.05} = 1.81$$

Decision Rule:

Reject H_0 if $F_{\text{STAT}} > F_{\text{CRITICAL}}$. Here, $34.62 > 1.81$

Since the computed F-statistic is far greater than the critical value, we reject H_0 at the 5 % significance level. This shows very strong statistical evidence that the independent variables are jointly significant in explaining Visa's stock price.

The extremely small p-value (p-value $< 2.2 \times 10^{-16}$) further supports this conclusion, as it is well below any conventional significance threshold. This outcome matches the high goodness-of-fit

measures, with an R^2 of 0.9499 and an adjusted R^2 of 0.9225, confirming that the full model provides substantial explanatory power.

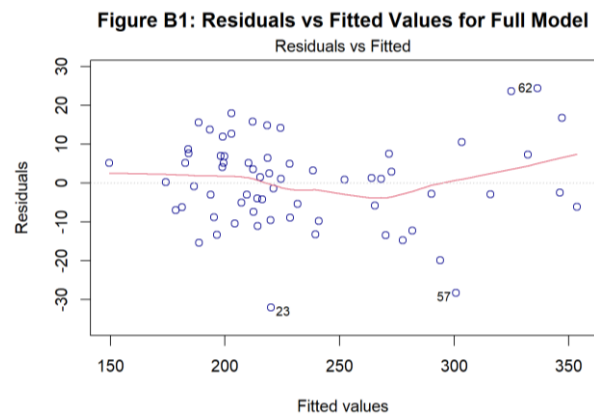
Having established both individual and joint significance, the next step is to examine whether the Gauss-Markov assumptions hold for the complete specification.

VI. Discussion of the Gauss-Markov assumptions for the Full regression model

A. Linearity in terms of the parameters

The full regression model specifies Visa's stock price as a linear function of twenty-three macroeconomic and financial variables. Since all predictors enter additively and with linear coefficients, the linearity-in-parameters condition is satisfied by design. The residuals vs fitted plot (see Figure B1) for the full model shows a mostly flat smoothing line with a small dip in the middle and a mild upward slope at higher fitted values. The residuals are spread around zero without a clear U-shape, strong curvature, or repeating pattern, indicating that the overall relationship between the regressors and Visa's price level aligns reasonably well with a linear functional form.

The Residual plots provide further confirmation. For most predictors, the residuals cluster around the fitted linear trend, suggesting that linearity is an appropriate approximation. While a few variables, such as CPI_Mom, Personal Income, and Consumer Loans, exhibit small departures from perfect linearity, these deviations are minor and do not necessitate nonlinear transformations. Together, the graphical diagnostics support the conclusion that the linearity assumption holds adequately for the full Visa regression model.



B. No Perfect Multicollinearity

The full Visa regression model was estimated without computational issues, meaning that none of the variables are perfectly collinear, and the assumption of no perfect multicollinearity is satisfied. However, the VIF results clearly show the presence of severe multicollinearity among many predictors. Several variables have extremely high VIF values, including PCE Consumption (around 2496), CPI Index (1336), Personal Income (924), Payroll Employment (231), Retail Sales (388), Real Disposable Income (148), Industrial Production (46), Unemployment Rate (56), and the Treasury yields Y10Y and Y2Y (94 and 123). These values far exceed standard thresholds, confirming that many macroeconomic indicators in the full model move together and contain overlapping information. While multicollinearity does not bias the coefficient estimates, it inflates standard errors and makes individual t-statistics unreliable, causing several economically meaningful variables to appear statistically insignificant. This diagnostic test highlights why the full model's coefficients are unstable and supports the shift toward a reduced five-variable specification, which substantially lowers collinearity and improves the interpretability and stability of the regression results. (*Belsley, Kuh & Welsch, 1980*)

C. Randomness and independence of error terms (No Autocorrelation)

A key Gauss-Markov requirement is that the error terms be independent across time. This means the residuals should not show systematic trends or temporal patterns when plotted. For the full model, this assumption is checked using both graphical diagnostics and formal statistical tests.

Graphical tools: The residuals versus fitted values plot shows a scattered pattern, but with noticeable clusters rather than a fully random spread. The Q-Q plot for the Visa model (Figure B1) shows that most points lie close to the theoretical normal line, although a few tail values deviate from perfect normality. Together, these plots suggest that the residuals are not completely random and may contain some time-related dependence.

Durbin-Watson test:

For further analysis, we perform the Durbin-Watson (DW) test to examine whether the residuals exhibit first-order autocorrelation. The underlying process can be written as:

$$\varepsilon_{i+1} = a \varepsilon_i + w_i \text{ (for } i = 1, 2, \dots, 66)$$

where:

a - deterministic coefficient; w_i - stochastic error term (noise); We test the following hypotheses:

$$H_0: a = 0 \text{ (no autocorrelation)}$$

$$H_1: a > 0 \text{ (positive autocorrelation)}$$

From the R output for the full model, the Durbin-Watson statistic is:

$$DW_{STAT} = 1.2157 \text{ with } p\text{-value} = 1.598 \times 10^{-6}$$

Decision Rule:

At the 5% significance level, we reject H_0 if the p-value < 0.05 . Since the p-value is far below 0.05, we reject the null hypothesis. This provides strong statistical evidence of positive first-order autocorrelation in the residuals. Such autocorrelation is common in models using monthly macroeconomic and financial variables, as these series tend to move gradually over time rather than behaving independently from one period to the next. (Durbin & Watson, 1950)

D. Zero Conditional Mean

OLS residuals should sum to zero. The R output confirms this condition, since the residuals sum to 1.03×10^{-14} , which is effectively zero aside from rounding. This supports the zero conditional mean assumption for the Visa full model. $\frac{1}{n} \sum_{t=1}^n \hat{\varepsilon}_t = 1.03 \times 10^{-14} \approx 0$, where $n = 66$

E. Homoskedasticity of errors

Homoskedasticity means that the variance of the error term is constant across all observations. If this assumption is violated, the model becomes heteroskedastic, so the OLS slope estimates remain unbiased, but the standard errors and t-statistics become unreliable. This assumption is checked using both graphical and formal tools.

Graphical tools: A visual inspection of Figure B1 (Residuals vs Fitted Values for VISA) shows the distribution of residuals across fitted values for the full regression model. If homoskedasticity holds, the residual cloud should display a horizontal band with constant spread. In the VISA plot, the spread of residuals widens slightly at higher fitted values, suggesting mild non-constant variance. Although the pattern is not severe, it suggests that some observations may exhibit greater variability than others, indicating that the homoskedasticity assumption may be partially weakened in the full 23-variable specification.

Breusch-Pagan test: To formally assess whether the variance of the error term depends on the regressors, the Breusch-Pagan (BP) test is applied. The BP test uses an auxiliary regression where the squared residuals are regressed on all explanatory variables. For the VISA model with 23 predictors, the auxiliary regression takes the form:

$$\hat{\varepsilon}_i^2 = b_0 + b_1 \text{VISA_Volume}_i + b_2 \text{SP500_Return}_i + \dots + b_{23} \text{Housing_Starts}_i + u_i.$$

In this context, $\hat{\varepsilon}_i$ represents the residuals from the full VISA regression model and is the error term of the auxiliary regression.

The hypotheses tested are as follows:

$$H_0: b_1 = b_2 = \dots = b_{23} = 0 \text{ (homoskedasticity)}$$

$$H_1: \text{not all } b_j = 0 \text{ (heteroskedasticity)}$$

The BP statistic is computed as:

$$BP = nR_{aux}^2 \sim \chi^2(k), \text{ where } n \text{ is the sample size and } k \text{ is the number of regressors.}$$

Where $k = 23$, $n = 66$

VISA Results: $BP_{STAT} = 22.818$, $p\text{-value} = 0.4714$

Since the p-value is greater than 0.05, we do not reject the null hypothesis, meaning there is no statistical evidence of general heteroskedasticity in the VISA full model. Although the residual plot shows a slight widening of variance at higher fitted values, both the Breusch–Pagan test and the White test report large p-values, indicating that these patterns are not statistically meaningful. Overall, the results suggest that the VISA full model does not suffer from significant heteroskedasticity, and the assumption of constant error variance is reasonably satisfied for this dataset. (*Breusch & Pagan, 1979*)

White test: To assess whether the full VISA regression model exhibits more general, nonlinear forms of heteroskedasticity, the White test is applied. This procedure involves estimating an auxiliary regression where the squared residuals from the original model, $\hat{\varepsilon}_i^2$, are regressed on the full set of explanatory variables, along with their squared terms and cross products. The hypotheses are:

$$H_0: \text{The Variance of } \varepsilon_i \text{ is constant}$$

$$H_1: \text{The Variance of } \varepsilon_i \text{ is not constant}$$

The test statistic is:

$$W = nR_{aux}^2$$

where $n = 66$, and the statistic follows a chi-square distribution with degrees of freedom equal to the number of regressors in the auxiliary regression (excluding the intercept).

VISA Results:

White Test Statistic = 4.24; $p\text{-value} = 0.119811$

Since the p-value exceeds the five per cent significance level, the null hypothesis is not rejected. The White test, therefore, provides no statistical evidence of general heteroskedasticity in the full VISA model. This suggests that the variance of the residuals does not show nonlinear dependence on the regressors when examined through the broader White-test framework. (*White, 1980*)

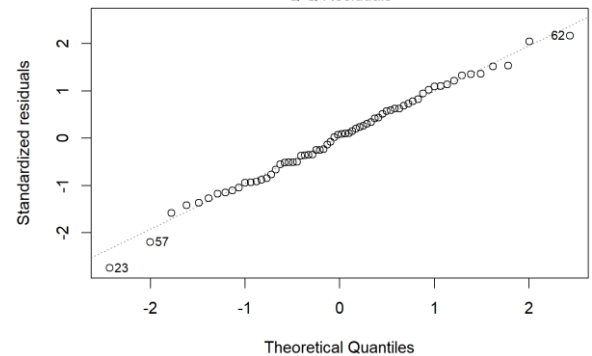
F. Normality of Error Terms

Checking whether the residuals follow a normal distribution helps determine if the standard t-tests, F-tests, and confidence intervals produced by the OLS model are reliable. Normality itself is not required for OLS to be unbiased, but it improves the accuracy of inference, so both visual and statistical methods are used to assess this assumption.

Graphical Check:

The Q-Q plot for the full VISA model (Figure B2) compares the standardised residuals with the theoretical quantiles of a normal distribution. Most points align closely with the reference line, indicating that the central part of the residual distribution follows the pattern expected under normality. Although a few observations drift away from the line at the upper and lower tails, these departures are modest and not unusual for monthly financial data. Taken together, the plot suggests that the normality assumption for the residuals is reasonably well satisfied, with no strong evidence of distortion that would undermine the regression results.

Figure B2: QQ Plot of Standardized Residuals for Full Model
Q-Q Residuals



VII. Interpretation and Treatment of Gauss-Markov Violations

The diagnostic checks for the full VISA regression model indicate that some of the classical OLS assumptions are not met perfectly. These violations do not render the model unusable, but they do impact the reliability of the standard errors and statistical tests. Below is a summary of each issue, along with the practical steps that can be taken to address them.

A. Autocorrelation in the residuals

The Durbin-Watson test gives $DW = 1.2157$ with a $p\text{-value} = 1.598e^{-6}$, showing strong positive autocorrelation. This means the errors follow a time-dependent pattern rather than occurring randomly. This is common in financial and macroeconomic data because many variables adjust slowly over time. A useful fix is to shift from a static regression to models that directly include lagged effects, such as AR(1), ARMA, ARIMA, or ARMA-GARCH. These models handle persistence more effectively and give more efficient estimates for time-dependent series.

B. Checking for Constant Error Variance (Homoskedasticity)

The residual plot shows a slight widening of residuals at higher fitted values, which suggests mild heteroskedasticity. However, both formal tests provide the opposite message:

Breusch-Pagan test: $BP = 22.818$; $p = 0.4714$

White test: $W = 4.24$; $p = 0.1198$

Since both tests report $p\text{-values}$ much greater than 0.05, the null hypothesis of constant variance cannot be rejected. This means there is no statistical evidence of heteroskedasticity in the full VISA model. Robust standard errors can still be applied as a safety measure, but they are not strictly required based on these results.

C. High Multicollinearity Among Predictors

The VIF results for VISA show extremely high values for many variables (for example, Retail_Sales, PCE_Consumption, CPI_Index, Y2Y, Payroll_Employment, Personal_Income). This means several predictors are highly correlated with each other. High multicollinearity inflates standard errors and makes coefficient signs and magnitudes unstable even when the variables are economically meaningful. The appropriate remedy is to remove or combine correlated predictors, keeping only those that capture unique economic information. This step led to the more stable five-variable VISA model used later in the analysis.

D. Normality of the Residuals

The Q-Q plot indicates that the residuals follow the theoretical normal line closely, with only small deviations at the tails. These minor departures are not large enough to threaten the normality assumption in practice, and OLS remains reliable in this setting. While perfect normality is not

essential, using robust standard errors provides additional protection if the residual distribution contains slight tail irregularities.

VIII. Reduction of Variables in the Regression Models.

A. Initial Variable Reduction: 25 Down to 11 Predictors

The full VISA model originally contained twenty-three predictors. Many of these variables showed high multicollinearity, weak individual t-statistics, or limited economic relevance when included together. The first reduction aims to remove predictors that add noise while keeping variables that are statistically stronger or economically meaningful for explaining movements in VISA's stock price.

The reduction was based on three criteria:

- (1) individual p-values and t-statistics from the full model,
- (2) Variance Inflation Factors (VIFs) to detect multicollinearity, and
- (3) economic relevance for a payments company such as VISA.

Variables that were consistently insignificant, unstable across different specifications, or highly correlated with other predictors were removed. Predictors with clearer economic relevance, such as consumer spending, labour-market strength, interest-rate conditions, and financial-market stress, were kept in the model.

The reduced model is estimated as:

$$\begin{aligned} \text{VISA_Price}_t = & \alpha_0 + \alpha_1 \cdot \text{SP500_Return}_t + \alpha_2 \cdot \text{Nasdaq_Return}_t + \alpha_3 \cdot \text{VIX_Level}_t + \alpha_4 \cdot \text{Oil_Price}_t \\ & + \alpha_5 \cdot \text{PCE_Consumption}_t + \alpha_6 \cdot \text{Unemp_Rate}_t + \alpha_7 \cdot \text{Y2Y}_t + \\ & + \alpha_8 \cdot \text{Financial_Conditions_Index}_t + \alpha_9 \cdot \text{Payroll_Employment}_t + \\ & + \alpha_{10} \cdot \text{Real_Disposable_Income}_t + \alpha_{11} \cdot \text{Consumer_Loans}_t + u_t \end{aligned}$$

where α_0 is the intercept representing the baseline level of Visa's price, and u_t is the error term capturing all influences not accounted for by the 11 predictors in the month t .

This reduced specification delivers an adjusted R^2 of 0.91, indicating that the model preserves most of the explanatory power of the full specification despite removing twelve variables. Several predictors, such as SP500_Return, PCE_Consumption, Unemp_Rate, and Payroll_Employment, show stronger t-statistics and smaller standard errors compared to the full model. This confirms that the reduced model improves coefficient precision, lowers the impact of multicollinearity, and provides a more stable and interpretable foundation for deriving the final model.

B. Final Reduction: From 11 to 5 Independent Variables

After estimating the 11-variable reduced model, a final refinement is carried out to keep only those predictors that show strong significance, economic relevance, and stable behaviour under diagnostic testing. The goal of this step is to produce a compact model that explains VISA's price movements without unnecessary multicollinearity or noise. Using the reduced model output, five variables consistently show strong explanatory power: PCE_Consumption, Unemp_Rate, Y2Y, Financial_Conditions_Index, and Payroll_Employment. These variables directly relate to consumer spending strength, labour market conditions, interest rate expectations, system-wide financial stress, and aggregate employment, all of which are fundamental drivers of transaction activity for a payments company like VISA.

The final regression specification is:

$$\text{VISA_Price}_t = \delta_0 + \delta_1 \cdot \text{PCE_Consumption}_t + \delta_2 \cdot \text{Unemp_Rate}_t + \delta_3 \cdot \text{Y2Y}_t + \delta_4 \cdot \text{Financial_Conditions_Index}_t + \delta_5 \cdot \text{Payroll_Employment}_t + e_t$$

where, δ_0 is the intercept showing the baseline level of Visa's price, and e_t is the error term capturing all influences not accounted for by the 5 predictors in the month t .

The final five-variable model shows strong statistical performance and provides a clear improvement over earlier specifications. All five coefficients are statistically significant at the 5% level, with t-statistics ranging from about 3.3 to 9.6, indicating that each variable contributes meaningfully to explaining VISA's price movements. The model achieves an R^2 of 0.9105 and an adjusted R^2 of 0.903, meaning that these predictors jointly explain more than 90% of the variation in monthly VISA stock prices. The overall F-statistic of 122.1 (with 5 and 60 degrees of freedom) and its p-value below 2.2×10^{-16} confirm that the model is highly significant as a group. Compared to the full model and the first reduction, this final specification delivers more stable coefficient estimates, reduced multicollinearity, clearer economic interpretation, and a better balance between goodness-of-fit and model simplicity.

IX. Discussion of Gauss Markov Assumptions for the Full Model

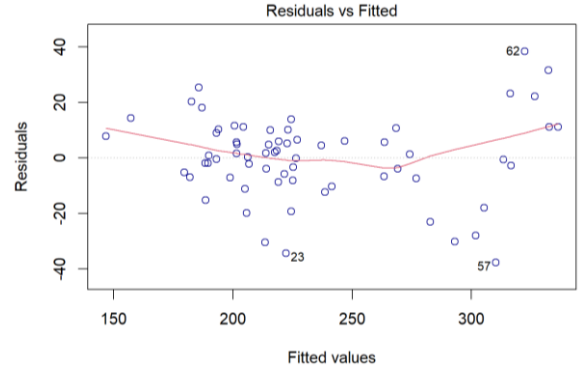
A. Linearity in terms of the parameters

The final VISA model expresses the stock price as a linear function of five macroeconomic predictors and an error term:

$$\text{VISA_Price}_t = \beta_0 + \beta_1 \cdot \text{PCE_Consumption}_t + \beta_2 \cdot \text{Unemp_Rate}_t + \beta_3 \cdot \text{Y2Y}_t + \beta_4 \cdot \text{Financial_Conditions_Index}_t + \beta_5 \cdot \text{Payroll_Employment}_t + \varepsilon_t$$

Because the specification is strictly additive and all coefficients enter linearly, the linearity-in-parameters assumption is satisfied. Graphical diagnostics also support this conclusion. The residuals-versus-fitted plot (Figure C1) shows a generally horizontal pattern with no distinct curvature, although some widening is visible at high fitted values. The component-plus-residual (CR) plots (Figure C3) indicate that the partial residuals for PCE Consumption, Unemployment Rate, and Financial Conditions Index follow linear trends closely. Minor curvature appears for Y2Y and Payroll Employment, but the magnitude is small and does not warrant nonlinear transformations. Overall, both the model specification and diagnostic plots confirm that the linearity assumption is reasonably met in the final VISA model.

Figure C1: Residuals vs Fitted Values for Final Model



B. No perfect Multicollinearity

The OLS estimates for the final model were successfully computed in R, confirming that the design matrix is of full rank and that perfect multicollinearity is absent. Variance Inflation Factors (VIFs) provide further insight into the degree of correlation among regressors:

PCE_Consumption: 19.32; Unemp_Rate: 6.55; Y2Y: 8.74; Financial_Conditions_Index: 2.37; Payroll_Employment: 29.29

None of the VIF values are infinite, so the assumption is satisfied. However, the results indicate moderate to high multicollinearity, particularly for PCE Consumption and Payroll Employment. This is expected because both variables reflect broader economic activity and tend to move together

with labour-market and income conditions. Despite the elevated VIFs, all coefficients remain statistically significant and removing them would weaken the economic interpretation of the model. Thus, while multicollinearity is present, it is not severe enough to undermine the final model.

C. Randomness and independence of error terms (No Autocorrelation)

Error independence requires that the residuals are not correlated across time. A visual inspection of the residual plot shows clustering and some trending behaviour, suggesting potential time dependence. To formally assess this, the Durbin-Watson test is conducted with the following specification:

$$\varepsilon_{t+1} = a \cdot \varepsilon_t + w_t$$

where a is a deterministic autoregressive coefficient and w_t is a random noise process.

We test:

$H_0: a = 0$ (no autocorrelation)

$H_1: a > 0$ (positive autocorrelation)

The output is: $DW_{STAT} = 0.901$, $p\text{-value} = 2.056 \times 10^{-8}$

Since the p -value is far below 0.05, the null is rejected. There is strong evidence of positive autocorrelation in the final model. This pattern is common in financial and macroeconomic series, where variables evolve gradually over time. The presence of autocorrelation does not bias OLS coefficients, but it does make standard errors unreliable. Dynamic models, such as AR, ARMA, or ARMA-GARCH, are better suited for capturing these time-dependent effects.

D. Zero Conditional Mean:

By the properties of OLS, the residuals satisfy $\frac{1}{n} \sum_{t=1}^n \hat{\varepsilon}_t = -1.296 \times 10^{-14} \approx 0$. With $n = 66$, allowing for minor numerical rounding, this confirms that the residuals average to zero, thereby satisfying the zero conditional mean assumption.

E. Homoskedasticity of errors

Homoskedasticity requires that the variance of the error term is constant across all observations. The residual-versus-fitted plot reveals a widening spread of residuals at higher fitted values, indicating potential heteroskedasticity. Formal tests confirm this visually observed pattern.

Breusch-Pagan Test:

$BP_{STAT} = 16.644$; $df = 5$; $p\text{-value} = 0.00523$

White Test:

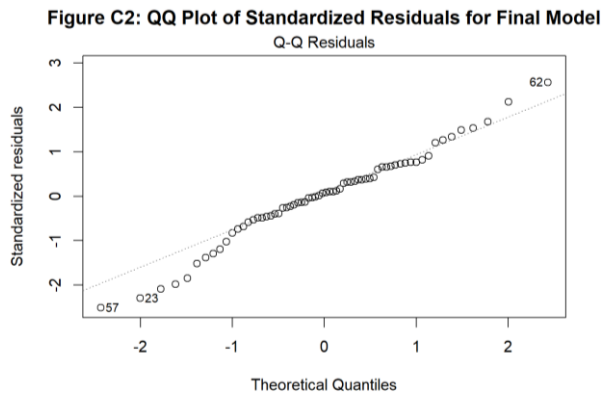
$W = 13.01$; $p\text{ value} = 0.00150$

Since both p -values are below 0.05, the null hypothesis of constant variance is rejected. The final model, therefore, exhibits statistically significant heteroskedasticity. While OLS coefficients remain unbiased, standard errors and t -statistics become unreliable. Heteroskedasticity-robust standard errors are recommended for proper inference.

F. Normality of Error Terms

The Q-Q plot of the standardised residuals (Figure C2) shows only mild deviations in the upper tail.

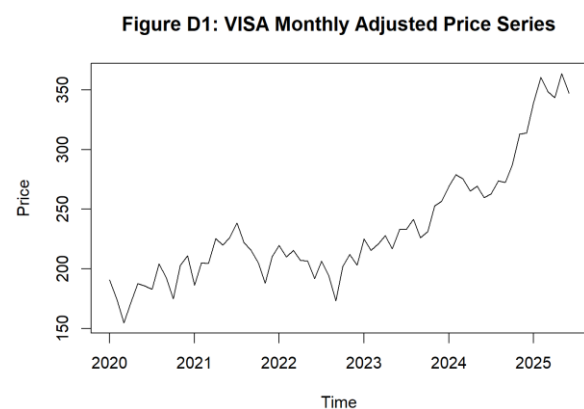
This indicates that the residuals are approximately normally distributed. Minor deviations in the tail are expected in financial data but are not severe enough to compromise inference, particularly with a sample of this size. Normality is therefore reasonably satisfied for the final model.



X. Statement and estimation of the time-series models

A. Stationarity Assessment of VISA Log Returns:

The plot of Visa's monthly adjusted price (Figure D1) shows that the share price increases steadily from early 2020 to mid-2025. The line does not move around a stable average but keeps rising, with short periods of dips and recoveries. From 2023 onwards, the price climbs at a faster pace, reaching its highest levels in 2024 and 2025. Because the series moves upward over time and does not return to any fixed level, it does not display the behaviour of a stationary series. This visual pattern suggests that the monthly price levels are non-stationary and that a transformation such as log returns is required before applying time-series models.



Augmented-Dickey-Fuller (ADF) Test

Hypotheses:

H_0 : The series contains a unit root.

H_1 : The series is not a unit root process.

Output:

Dickey-Fuller statistic (ADF Statistic) = -1.2593; p value = 0.8756

Interpretation:

The p-value is much larger than the conventional significance level of 0.05. Therefore, the null hypothesis cannot be rejected. This indicates that the VISA price series exhibits unit root behaviour. In other words, it is non-stationary in levels. (Dickey & Fuller, 1979)

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test

Hypotheses:

H_0 : The series is stationary around a level.

H_1 : The series is non-stationary.

Output:

KPSS statistic = 1.358; p value = 0.01

Interpretation:

The p-value is below 0.05. This leads to rejection of the null hypothesis of stationarity. The result suggests that the VISA price series is non-stationary. (Kwiatkowski et al., 1992)

Both tests point in the same direction.

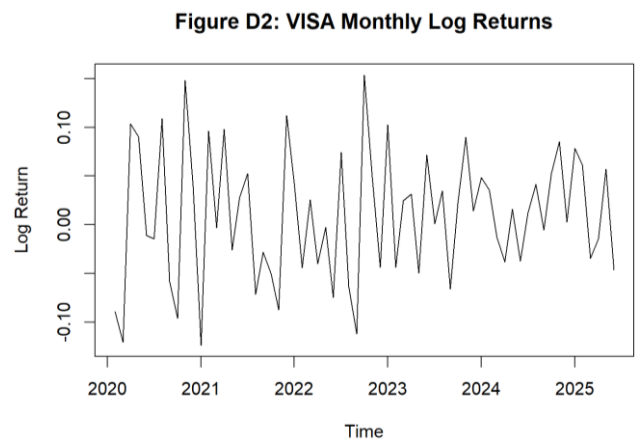
The ADF test fails to reject the presence of a unit root.

The KPSS test rejects level stationarity.

Taken together, the evidence supports that the VISA price series is non-stationary in levels. This outcome aligns with typical financial time series, where prices trend over time. As a result, transforming the series into log returns is appropriate. The transformation removes the trend and stabilises the variance, making the resulting series more suitable for building time series models such as ARIMA or GARCH. (Kwiatkowski *et al.*, 1992)

B. Stationarity Tests for VISA Log Return Series:

After transforming the VISA price series into monthly log returns, the behaviour of the data becomes more stable. The plot of the log returns shows irregular fluctuations around a roughly constant mean, with no visible upward or downward trend. The amplitude of the variations also appears stable over time, indicating that the transformation has removed the trend component that was present in the price series. To confirm the suitability of this series for time series modelling, formal stationarity tests are conducted. (Tsay, 2010; Engle, 1982; Bollerslev, 1986)



Augmented-Dickey-Fuller (ADF) Test

Hypotheses:

H_0 : The series contains a unit root.

H_1 : The series is stationary.

Output:

Dickey Fuller statistic = -4.2315, p value = 0.01

Interpretation:

The p-value is below 0.05. The null hypothesis of a unit root is rejected. This indicates that the VISA log return series is stationary. The result is consistent with the behaviour expected of financial return series, which typically fluctuate around a stable long-term mean once the underlying price trend has been removed.

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test

Hypotheses:

H_0 : The series is stationary.

H_1 : The series is non-stationary.

Output:

KPSS statistic = 0.13948, p value = 0.1

Interpretation:

Since the p-value is greater than 0.05, there is no evidence against the null hypothesis. The KPSS result supports the conclusion that the log return series is level stationary. (Kwiatkowski *et al.*, 1992)

Combined Conclusion:

The ADF and KPSS results show that the VISA log return series is stationary. The ADF test rejects a unit root, and the KPSS test does not reject stationarity. This confirms that the log return transformation creates a stable series suitable for time series models.

C. AR (1) Model:

The AR(1) structure serves as a fundamental baseline for modelling short-term dependence in the VISA log returns. The autoregressive coefficient is estimated at approximately -0.2137 with a standard error of 0.1231. Although the sign suggests mild negative dependence, the coefficient is not statistically significant. This indicates that past monthly returns do not exert a reliable influence on current movements. The intercept is approximately 0.0096, which implies an average monthly return of around 0.96%. The estimated innovation variance is 0.004173, and the associated AIC is -165.64. These values confirm that the series is mostly driven by random shocks rather than persistent autoregressive effects. The AR(1) specification, therefore, captures the broad randomness of VISA returns but does not uncover strong predictability.

D. ARMA (1,1) Model:

The ARMA(1,1) model introduces both an autoregressive and a moving average term to capture more complex short-run interactions. The AR coefficient is estimated at 0.3431 with a standard error of 0.2920, while the MA coefficient is -0.6671 with a standard error of 0.2295. The relatively large standard errors reduce the statistical reliability of these coefficients, although the negative MA term is more precisely estimated. The innovation variance decreases slightly to 0.003918, and the log likelihood improves to 87.79. The AIC also improves to -167.58, making this specification marginally preferable to the AR(1) model. However, the improvement in fit is modest, and the coefficient estimates do not suggest strong or stable dynamics in the conditional mean. This outcome implies that adding a moving average term provides only a limited benefit in modelling monthly VISA returns.

E. ARIMA (1,1,1) Model:

The ARIMA(1,1,1) specification is estimated to test whether differencing the log returns enhances the model fit. Since the original log return series is already stationary, differencing introduces unnecessary noise, which is evident from the parameter behaviour. The MA coefficient reaches -1.000 with a very small standard error, indicating a boundary estimate rather than a meaningful relationship. The AR coefficient is -0.2008 with a standard error of 0.1247 and remains statistically insignificant. The estimated variance rises to 0.004239, the log likelihood drops to 81.73, and the AIC rises to -157.46. These results confirm that differencing an already stationary return series degrades model performance and produces unstable or near-degenerate parameter estimates. The ARIMA structure, therefore, offers no modelling advantages for VISA returns.

F. ARMA (1,1)-GARCH (1,1) Model:

The ARMA(1,1)-GARCH(1,1) model gives the most complete picture of how VISA monthly returns behave. In the mean equation, the intercept is about 0.0094, and the MA coefficient (-0.5443) is statistically significant. This means that once we allow volatility to change over time, the short-term movements in returns are captured more reliably. The AR term is not very precise, which is normal for stock return data. In the variance equation, the GARCH coefficient is 0.9168, and the ARCH coefficient is 0.0523. These values show that today's volatility depends mostly on past volatility, with a smaller effect from recent shocks. The constant term is close to zero, indicating a

low long-run variance. The log likelihood increases to 90.49, the highest among all models, and the information criteria also improve. The residual tests indicate no serial correlation, no leftover ARCH effects, and acceptable normality, suggesting that the model is well-specified. Overall, these results show that simpler AR or ARMA models cannot fully capture the behaviour of VISA returns, while the ARMA-GARCH model handles both weak return dependence and strong volatility persistence, making it the best model for explaining and forecasting VISA log returns. (*Engle, 1982; Bollerslev, 1986*)

G. Comparison of the Time Series Models

The forecasted SSR values are all very small, indicating that the log-return models are able to predict short-horizon movements with good accuracy. The AR(1) model produces the lowest SSR at 0.026959, making it the best performer in this comparison. The ARMA(1,1) and ARIMA(1,1,1) models follow very closely, with SSR values of 0.026984 and 0.026975, showing almost no meaningful improvement from adding additional autoregressive or moving-average terms. The ARMA(1,1) model with a GARCH(1,1) volatility structure records a slightly higher SSR of 0.027021, indicating that modelling changing volatility does not materially enhance forecast accuracy for Visa's monthly log returns. Taken together, the results show that all four models forecast similarly well, but the simple AR(1) specification provides the most accurate out-of-sample predictions for this dataset.

Conclusion:

This study analysed Visa's monthly data using regression models in price levels and time-series models in log returns. The full regression model showed a high adjusted R^2 of about 0.92, suggesting that broad economic conditions are linked to movements in Visa's share price. However, the model had a large SSR of around 8148 and suffered from serious statistical problems. Multicollinearity was substantial, with several variables showing extremely high VIF values, and the Durbin-Watson test indicated clear positive autocorrelation in the residuals. As the model was reduced to fewer variables, the SSR increased further, reaching about 14,555 in the final five-variable specification, showing that monthly price levels are difficult to explain reliably with macroeconomic indicators alone.

Switching to log returns produced a more stable and statistically suitable framework. The price series was non-stationary, while the return series passed both the ADF and KPSS tests, confirming stationarity. The AR(1), ARMA(1,1), ARIMA(1,1,1), and ARMA-GARCH models all delivered very small forecast errors, with SSR values close to 0.027. These results show that Visa's short-run behaviour is better captured through time-series methods than by regressions in levels. The similar performance across the models also suggests that the return series contains only limited persistence. In summary, the regression models help identify the broader economic forces associated with Visa's valuation, but they face significant statistical limitations when applied to price levels. Time-series models applied to stationary log returns provide a far clearer and more reliable description of Visa's short-term return movements, making them the more appropriate approach for prediction and modelling.

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XI. Statement and estimation of the time-series models

A. Full Model:

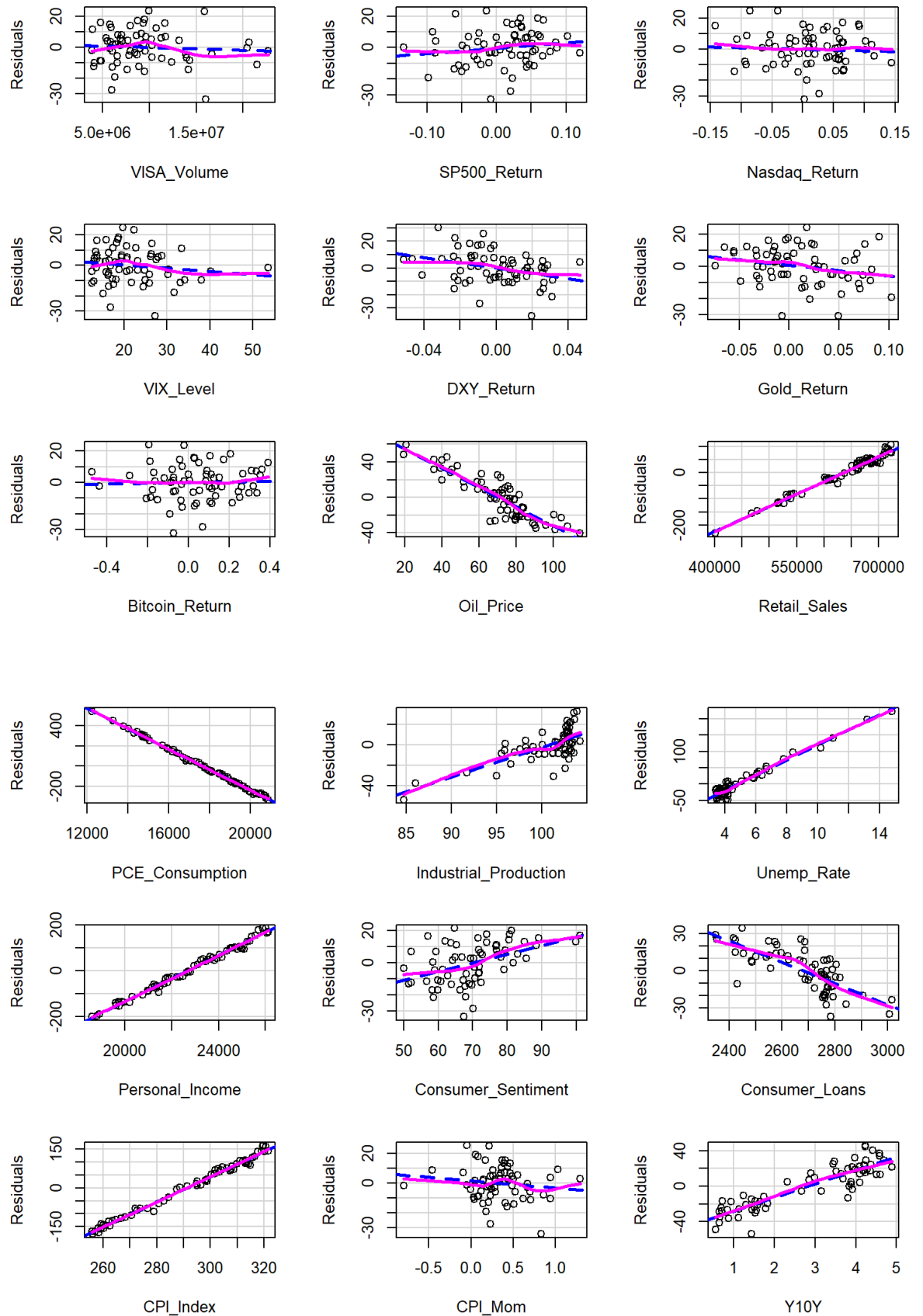
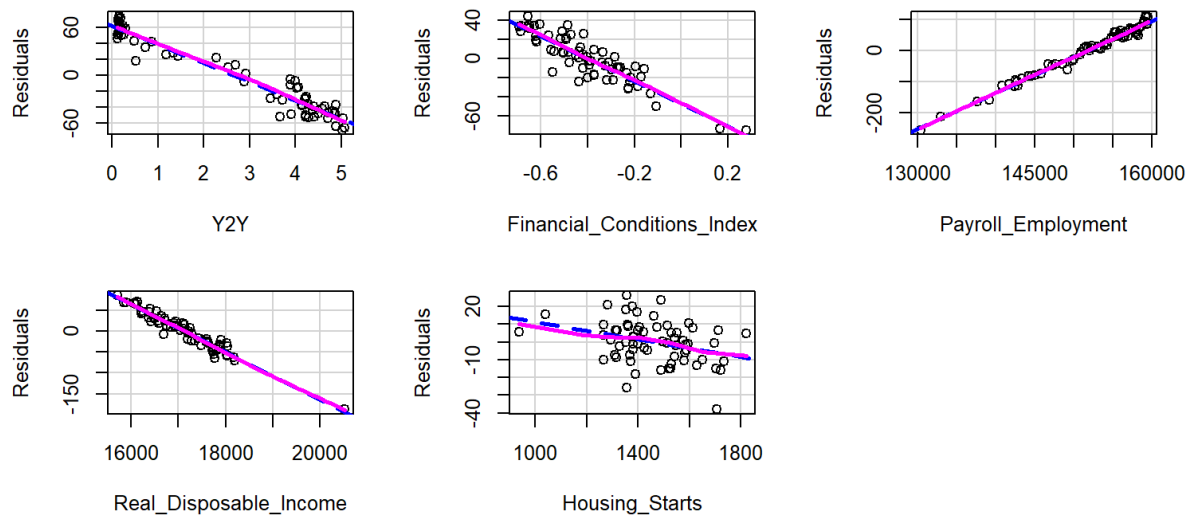
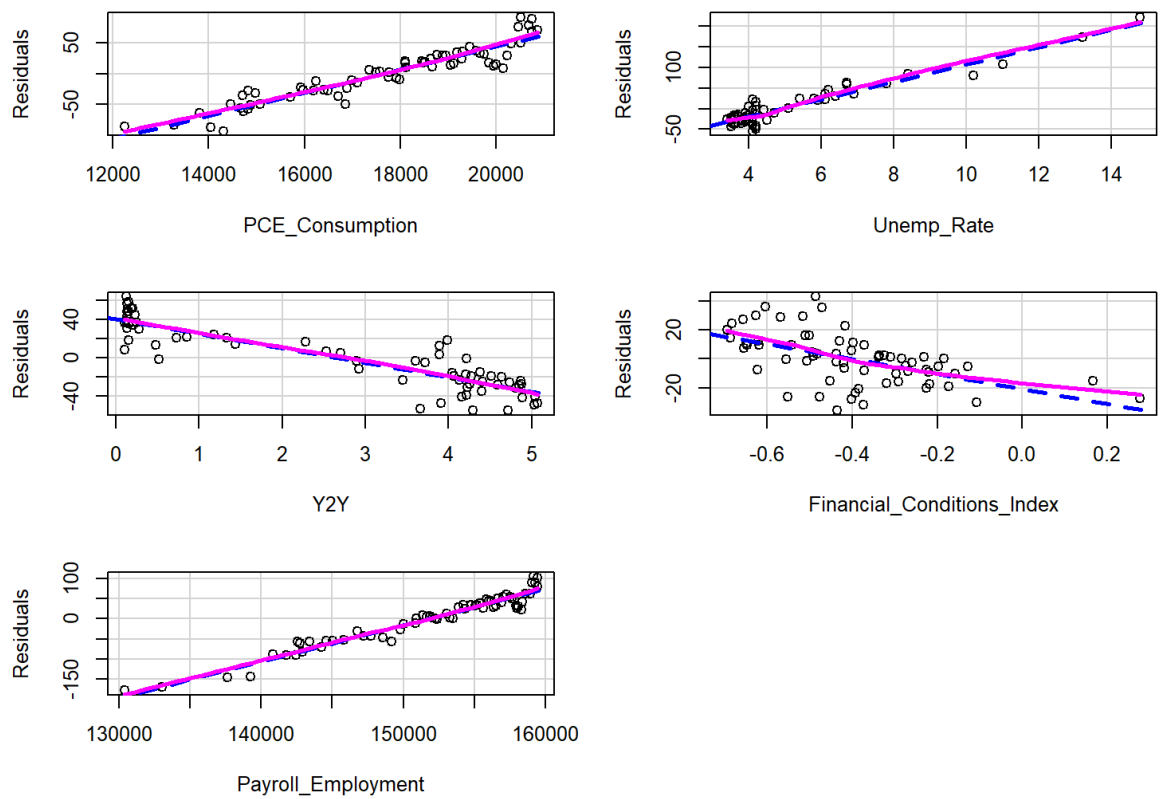


Figure B3: Residual Plots for all independent variables



B. Final Model:

Figure C3: Component plus Residual Plots for Final Model



C. Preprocessing Code:

```
#####
# A) REGRESSION SECTION - VISA
#####

#####
# 1. Load packages
#####

library(quantmod)

## Warning: package 'quantmod' was built under R version 4.5.2

## Loading required package: xts

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric

## Loading required package: TTR

## Warning: package 'TTR' was built under R version 4.5.2

## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo

library(tidyverse)

## — Attaching core tidyverse packages ————— tidyverse
2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.1      ✓ stringr    1.5.2
## ✓ ggplot2     4.0.0      ✓ tibble     3.3.0
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.1.0

## — Conflicts ————— tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::first()  masks xts::first()
## ✗ dplyr::lag()    masks stats::lag()
## ✗ dplyr::last()   masks xts::last()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors

library(GGally)
library(lubridate)
library(zoo)
```

```
library(xts)
library(car)

## Warning: package 'car' was built under R version 4.5.2

## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some

library(lmtest)

## Warning: package 'lmtest' was built under R version 4.5.2

library(whitestrapp)

## Warning: package 'whitestrapp' was built under R version 4.5.2
##
## Please cite as:
##
## Lopez, J. (2020), White's test and Bootstrapped White's test under the
methodology of Jeong, J., Lee, K. (1999) package version 0.0.1

library(memisc)

## Loading required package: lattice
## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
##
## Attaching package: 'memisc'
##
## The following object is masked from 'package:car':
##
##     recode
##
## The following objects are masked from 'package:lubridate':
##
##     as.interval, is.interval
##
## The following objects are masked from 'package:dplyr':
##
##     collect, recode, rename, syms
```

```
##
## The following object is masked from 'package:purrr':
##
##    %@%
##
## The following object is masked from 'package:tibble':
##
##    view
##
## The following object is masked from 'package:ggplot2':
##
##    syms
##
## The following objects are masked from 'package:stats':
##
##    contr.sum, contr.treatment, contrasts
##
## The following object is masked from 'package:base':
##
##    as.array

library(tseries)

## Warning: package 'tseries' was built under R version 4.5.2

library(fGarch)

## Warning: package 'fGarch' was built under R version 4.5.2

## NOTE: Packages 'fBasics', 'timeDate', and 'timeSeries' are no longer
## attached to the search() path when 'fGarch' is attached.
##
## If needed attach them yourself in your R script by e.g.,
##     require("timeSeries")
##
## Attaching package: 'fGarch'
##
## The following object is masked from 'package:TTR':
##
##    volatility

library(openxlsx)

## Warning: package 'openxlsx' was built under R version 4.5.2

library(sandwich)

## Warning: package 'sandwich' was built under R version 4.5.2

library(corrplot)

## Warning: package 'corrplot' was built under R version 4.5.2

## corrplot 0.95 loaded

library(dplyr)
```

```
#####
# 2. Download Data (Yahoo Finance + FRED)
#####

start_date <- as.Date("2019-12-01")
end_date   <- as.Date("2025-06-30")

# Yahoo Finance tickers for Visa and market/asset factors
tickers <- c("V", "^GSPC", "^IXIC", "^VIX", "DX-Y.NYB", "GLD", "BTC-USD")
getSymbols(tickers, src = "yahoo", from = start_date, to = end_date)

## Warning: ^VIX contains missing values. Some functions will not work if
objects
## contain missing values in the middle of the series. Consider using na.o
mit(),
## na.approx(), na.fill(), etc to remove or replace them.

## Warning: DX-Y.NYB contains missing values. Some functions will not work
if
## objects contain missing values in the middle of the series. Consider us
ing
## na.omit(), na.approx(), na.fill(), etc to remove or replace them.

## [1] "V"          "GSPC"        "IXIC"        "VIX"         "DX-Y.NYB"    "GLD"        "
BTC-USD"

# FRED series
fred_codes <- c(
  "RSAFS",      # Retail sales
  "PCE",        # Personal consumption expenditures
  "INDPRO",     # Industrial production
  "UNRATE",     # Unemployment rate
  "CPIAUCSL",   # CPI index
  "DGS10",      # 10 year Treasury
  "DGS2",       # 2 year Treasury
  "TOTCI",      # Consumer Loans
  "NFCI",       # National Financial Conditions Index
  "PI",         # Personal income
  "UMCSENT",    # Consumer sentiment
  "DCOILWTICO", # WTI oil
  "PAYEMS",     # Payroll employment
  "DSPIC96",    # Real disposable income
  "HOUST"      # Housing starts
)

getSymbols(fred_codes, src = "FRED", from = start_date, to = end_date)

## [1] "RSAFS"      "PCE"        "INDPRO"     "UNRATE"     "CPIAUCSL"
## [6] "DGS10"      "DGS2"       "TOTCI"      "NFCI"       "PI"
## [11] "UMCSENT"    "DCOILWTICO" "PAYEMS"     "DSPIC96"    "HOUST"

#####
# 3. Convert to Monthly
#####
```

```

# Visa and market prices (adjusted close monthly)
V_m      <- to.monthly(Ad(V),           indexAt = "lastof", OHLC = FALSE)
SPX_m    <- to.monthly(Ad(GSPC),        indexAt = "lastof", OHLC = FALSE)
IXIC_m   <- to.monthly(Ad(IXIC),        indexAt = "lastof", OHLC = FALSE)
VIX_m    <- to.monthly(Ad(VIX),         indexAt = "lastof", OHLC = FALSE)

## Warning in to.period(x, "months", indexAt = indexAt, name = name, ...):
## missing
## values removed from data

DXY_m    <- to.monthly(Ad(`DX-Y.NYB`),  indexAt = "lastof", OHLC = FALSE)

## Warning in to.period(x, "months", indexAt = indexAt, name = name, ...):
## missing
## values removed from data

GLD_m    <- to.monthly(Ad(GLD),          indexAt = "lastof", OHLC = FALSE)
BTC_m    <- to.monthly(Ad(`BTC-USD`),    indexAt = "lastof", OHLC = FALSE)

# Rename
colnames(V_m)      <- "VISA_Price"
colnames(SPX_m)    <- "SPX_Level"
colnames(IXIC_m)   <- "Nasdaq_Level"
colnames(VIX_m)    <- "VIX_Level"
colnames(DXY_m)    <- "DXY_Level"
colnames(GLD_m)    <- "Gold_Level"
colnames(BTC_m)    <- "BTC_Level"

# Visa trading volume (monthly)
V_vol_m <- to.monthly(Vo(V), indexAt = "lastof", OHLC = FALSE)
colnames(V_vol_m) <- "VISA_Volume"

# FRED monthly series
Retail_m <- to.monthly(RSAFS,          indexAt = "lastof", OHLC = FALSE)
PCE_m    <- to.monthly(PCE,            indexAt = "lastof", OHLC = FALSE)
IND_m    <- to.monthly(INDPRO,          indexAt = "lastof", OHLC = FALSE)
UNR_m    <- to.monthly(UNRATE,          indexAt = "lastof", OHLC = FALSE)
CPI_m    <- to.monthly(CPIAUCSL,        indexAt = "lastof", OHLC = FALSE)
Y10_m    <- to.monthly(DGS10,           indexAt = "lastof", OHLC = FALSE)

## Warning in to.period(x, "months", indexAt = indexAt, name = name, ...):
## missing
## values removed from data

Y2_m      <- to.monthly(DGS2,           indexAt = "lastof", OHLC = FALSE)

## Warning in to.period(x, "months", indexAt = indexAt, name = name, ...):
## missing
## values removed from data

Loans_m   <- to.monthly(TOTCI,          indexAt = "lastof", OHLC = FALSE)
NFCI_m    <- to.monthly(NFCI,           indexAt = "lastof", OHLC = FALSE)
Income_m  <- to.monthly(PI,             indexAt = "lastof", OHLC = FALSE)
Sent_m    <- to.monthly(UMCSENT,        indexAt = "lastof", OHLC = FALSE)
Oil_m     <- to.monthly(DCOILWTICO,     indexAt = "lastof", OHLC = FALSE)

```

```

## Warning in to.period(x, "months", indexAt = indexAt, name = name, ...):
missing
## values removed from data

Payroll_m <- to.monthly(PAYEMS, indexAt = "lastof", OHLC = FALSE)
RDIncome_m <- to.monthly(DSPIC96, indexAt = "lastof", OHLC = FALSE)
Housing_m <- to.monthly(HOUST, indexAt = "lastof", OHLC = FALSE)

#####
# 4. Derived Variables
#####

# Market and asset Log returns
SPX_ret_raw <- diff(log(SPX_m))
Nasdaq_ret_raw <- diff(log(IXIC_m))
DXY_ret_raw <- diff(log(DXY_m))
Gold_ret_raw <- diff(log(GLD_m))
BTC_ret_raw <- diff(log(BTC_m))

# CPI month over month (percent)
CPI_Mom_raw <- diff(log(CPI_m)) * 100

# Rename derived series
colnames(SPX_ret_raw) <- "SP500_Return"
colnames(Nasdaq_ret_raw) <- "Nasdaq_Return"
colnames(DXY_ret_raw) <- "DXY_Return"
colnames(Gold_ret_raw) <- "Gold_Return"
colnames(BTC_ret_raw) <- "Bitcoin_Return"
colnames(CPI_Mom_raw) <- "CPI_Mom"

#####
# 5. Master Dataset Construction and Correlation Heatmap Analysis
#####
# 5A. Merge All Data to a Master Data Set
df_raw <- merge(
  V_m, V_vol_m,
  SPX_ret_raw, Nasdaq_ret_raw, VIX_m, DXY_ret_raw,
  Gold_ret_raw, BTC_ret_raw,
  Oil_m,
  Retail_m, PCE_m, IND_m, UNR_m,
  Income_m, Sent_m, Loans_m,
  CPI_m, CPI_Mom_raw, Y10_m, Y2_m,
  NFCI_m,
  Payroll_m, RDIncome_m, Housing_m
)

colnames(df_raw) <- c(
  "VISA_Price", "VISA_Volume",
  "SP500_Return", "Nasdaq_Return", "VIX_Level", "DXY_Return",
  "Gold_Return", "Bitcoin_Return",
  "Oil_Price",
  "Retail_Sales", "PCE_Consumption", "Industrial_Production", "Unemp_Rate",
  "Personal_Income", "Consumer_Sentiment", "Consumer_Loans",

```

```

"CPI_Index", "CPI_Mom", "Y10Y", "Y2Y",
"Financial_Conditions_Index",
"Payroll_Employment", "Real_Disposable_Income", "Housing_Starts"
)

df_xts <- na.omit(df_raw)
df <- data.frame(date = index(df_xts), coredata(df_xts))

```

D. Final Code:

```

#####
# A) MULTIVARIATE LINEAR REGRESSION
#####

# 1. Load packages
library(car)

## Warning: package 'car' was built under R version 4.5.2
## Loading required package: carData

library(lmtest)

## Warning: package 'lmtest' was built under R version 4.5.2
## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric

library(whitestrapp)

## Warning: package 'whitestrapp' was built under R version 4.5.2
##
## Please cite as:
##
## Lopez, J. (2020), White's test and Bootstrapped White's test under the
## methodology of Jeong, J., Lee, K. (1999) package version 0.0.1

library(memisc)

## Loading required package: lattice
## Loading required package: MASS

##
## Attaching package: 'memisc'

## The following object is masked from 'package:car':
##
##   recode

```

```

## The following objects are masked from 'package:stats':
##
##   contr.sum, contr.treatment, contrasts

## The following object is masked from 'package:base':
##
##   as.array

library(tseries)

## Warning: package 'tseries' was built under R version 4.5.2

## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo

library(fGarch)

## Warning: package 'fGarch' was built under R version 4.5.2

## NOTE: Packages 'fBasics', 'timeDate', and 'timeSeries' are no longer
## attached to the search() path when 'fGarch' is attached.
##
## If needed attach them yourself in your R script by e.g.,
##   require("timeSeries")

library(openxlsx)

## Warning: package 'openxlsx' was built under R version 4.5.2

library(corrplot)

## Warning: package 'corrplot' was built under R version 4.5.2

## corrplot 0.95 loaded

library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:memisc':
##
##   collect, recode, rename, syms

## The following object is masked from 'package:MASS':
##
##   select

## The following object is masked from 'package:car':
##
##   recode

## The following objects are masked from 'package:stats':
##
##   filter, lag

```

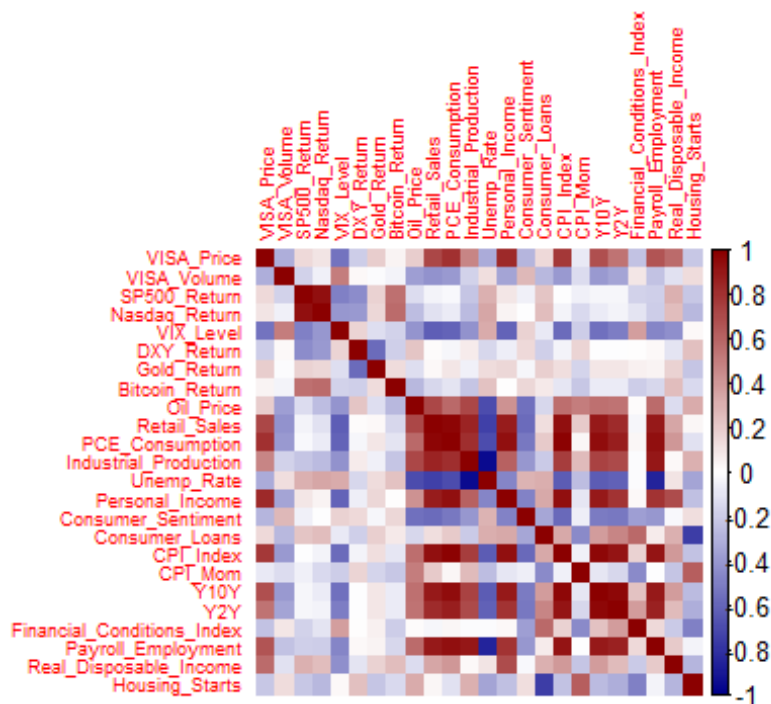
```
## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union

# 2. Read the Excel file
df <- as.data.frame(read.xlsx("VISA_Final_Dataset_Clean.xlsx"))
df$date <- as.Date(df$date)

# 3. Correlation Heatmap
df_num <- df %>% dplyr::select(where(is.numeric))
corr_matrix <- cor(df_num)

corrplot(
  corr_matrix,
  method = "color",
  type = "full",
  tl.cex = 0.55,
  col = colorRampPalette(c("darkblue", "white", "darkred"))(200),
  title = "Figure A: Correlation Heatmap of VISA Predictors",
  mar = c(0,0,2,0)
)
```

Figure A: Correlation Heatmap of VISA Predictors



```
# 4. Full OLS Model
model_full <- lm(
  VISA_Price ~ VISA_Volume +
    SP500_Return + Nasdaq_Return + VIX_Level +
    DXY_Return + Gold_Return + Bitcoin_Return +
    Oil_Price + Retail_Sales + PCE_Consumption + Industrial_Production +
    Unemp_Rate + Personal_Income +
    Consumer_Sentiment + Consumer_Loans +
    CPI_Index + CPI_Mom +

```

```

Y10Y + Y2Y +
Financial_Conditions_Index +
Payroll_Employment + Real_Disposable_Income +
Housing_Starts,
data = df
)

coef(model_full)

##              (Intercept)              VISA_Volume
##      -2.054746e+03      -1.867286e-07
##      SP500_Return      Nasdaq_Return
##      3.508937e+01      -1.078301e+01
##      VIX_Level      DXY_Return
##      -2.084348e-01      -1.986339e+02
##      Gold_Return      Bitcoin_Return
##      -6.640399e+01      2.011355e+00
##      Oil_Price      Retail_Sales
##      -1.084558e+00      9.550045e-04
##      PCE_Consumption      Industrial_Production
##      -1.025521e-01      2.894655e+00
##      Unemp_Rate      Personal_Income
##      2.319627e+01      5.046775e-02
##      Consumer_Sentiment      Consumer_Loans
##      5.428280e-01      -8.563790e-02
##      CPI_Index      CPI_Mom
##      4.897306e+00      -4.725767e+00
##      Y10Y      Y2Y
##      1.534336e+01      -2.327680e+01
## Financial_Conditions_Index      Payroll_Employment
##      -1.176474e+02      1.166152e-02
##      Real_Disposable_Income      Housing_Starts
##      -5.663582e-02      -2.456961e-02

summary(model_full)

##
## Call:
## lm(formula = VISA_Price ~ VISA_Volume + SP500_Return + Nasdaq_Return +
##      VIX_Level + DXY_Return + Gold_Return + Bitcoin_Return + Oil_Price +
##      Retail_Sales + PCE_Consumption + Industrial_Production +
##      Unemp_Rate + Personal_Income + Consumer_Sentiment + Consumer_Loans
##      +
##      CPI_Index + CPI_Mom + Y10Y + Y2Y + Financial_Conditions_Index +
##      Payroll_Employment + Real_Disposable_Income + Housing_Starts,
##      data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -32.109  -6.861   0.929   6.912  24.398
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2.055e+03  8.347e+02  -2.462 0.018016 *

```

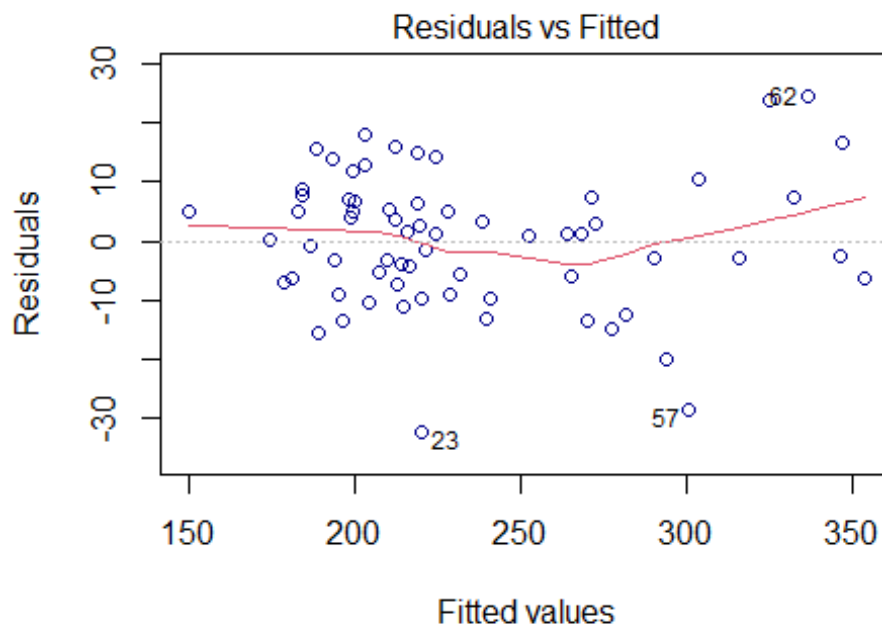
```

## VISA_Volume -1.867e-07 6.539e-07 -0.286 0.776630
## SP500_Return 3.509e+01 1.492e+02 0.235 0.815188
## Nasdaq_Return -1.078e+01 1.138e+02 -0.095 0.924949
## VIX_Level -2.084e-01 7.541e-01 -0.276 0.783600
## DXY_Return -1.986e+02 1.432e+02 -1.387 0.172821
## Gold_Return -6.640e+01 5.795e+01 -1.146 0.258367
## Bitcoin_Return 2.011e+00 1.474e+01 0.136 0.892100
## Oil_Price -1.085e+00 3.387e-01 -3.202 0.002605 **
## Retail_Sales 9.550e-04 4.646e-04 2.056 0.046072 *
## PCE_Consumption -1.026e-01 3.916e-02 -2.619 0.012214 *
## Industrial_Production 2.895e+00 3.097e+00 0.935 0.355258
## Unemp_Rate 2.320e+01 5.755e+00 4.031 0.000229 ***
## Personal_Income 5.047e-02 2.515e-02 2.006 0.051273 .
## Consumer_Sentiment 5.428e-01 4.502e-01 1.206 0.234616
## Consumer_Loans -8.564e-02 4.442e-02 -1.928 0.060635 .
## CPI_Index 4.897e+00 2.860e+00 1.713 0.094175 .
## CPI_Mom -4.726e+00 1.013e+01 -0.466 0.643311
## Y10Y 1.534e+01 1.168e+01 1.314 0.196041
## Y2Y -2.328e+01 9.867e+00 -2.359 0.023042 *
## Financial_Conditions_Index -1.176e+02 3.746e+01 -3.141 0.003085 **
## Payroll_Employment 1.166e-02 3.806e-03 3.064 0.003807 **
## Real_Disposable_Income -5.664e-02 2.677e-02 -2.115 0.040376 *
## Housing_Starts -2.457e-02 2.620e-02 -0.938 0.353738
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.93 on 42 degrees of freedom
## Multiple R-squared: 0.9499, Adjusted R-squared: 0.9225
## F-statistic: 34.62 on 23 and 42 DF, p-value: < 2.2e-16

# 5. Checking the Gauss Markov assumptions
plot(model_full, which = 1, col = "darkblue", sub = "",
      main = "Figure B1: Residuals vs Fitted Values for Full Model")

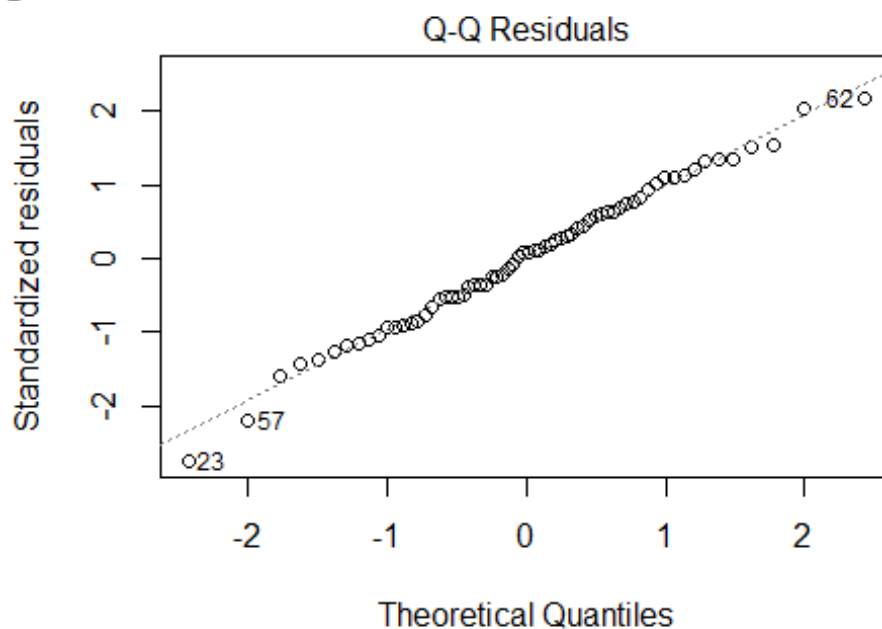
```

Figure B1: Residuals vs Fitted Values for Full Mod



```
plot(model_full, which = 2, col = "black", sub = "",  
     main = "Figure B2: QQ Plot of Standardized Residuals for Full Model")
```

Figure B2: QQ Plot of Standardized Residuals for Full Model



```
vif(model_full)
```

##	VISA_Volume	SP500_Return
##	2.713386	19.803564
##	Nasdaq_Return	VIX_Level

```
##          16.183961          11.285146
##          DXY_Return          Gold_Return
##          2.856630          1.915767
##          Bitcoin_Return          Oil_Price
##          2.498371          14.483199
##          Retail_Sales          PCE_Consumption
##          388.171852          2496.435703
##          Industrial_Production          Unemp_Rate
##          46.412921          56.235228
##          Personal_Income          Consumer_Sentiment
##          924.542974          7.648994
##          Consumer_Loans          CPI_Index
##          13.954809          1336.471797
##          CPI_Mom          Y10Y
##          3.610315          94.477865
##          Y2Y_Financial_Conditions_Index
##          123.316897          17.020580
##          Payroll_Employment          Real_Disposable_Income
##          231.175835          148.494638
##          Housing_Starts
##          5.804386

car::crPlots(model_full, ylab = "Residuals",
             main = "Figure B3: Residual Plots for all independent variables")
```

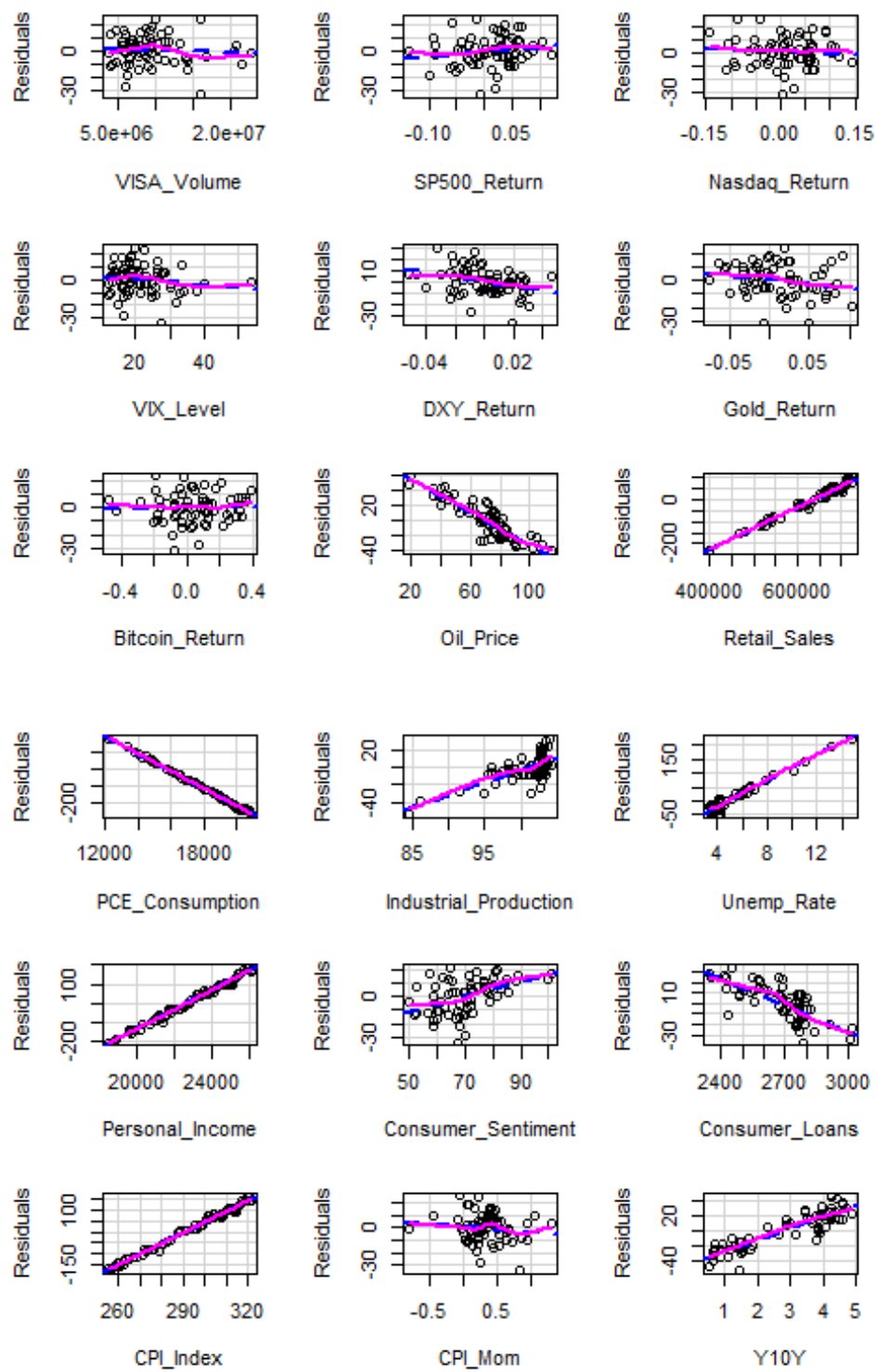
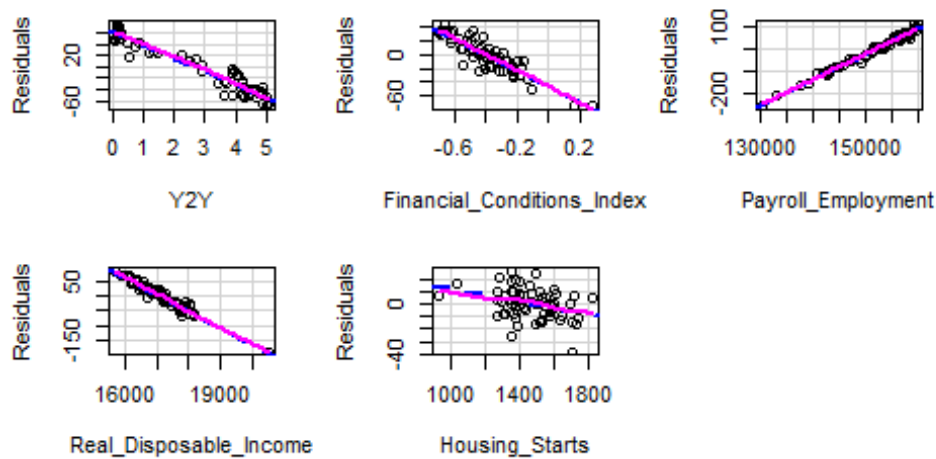


Figure B3: Residual Plots for all independent variables



```
dwtest(model_full)

##
## Durbin-Watson test
##
## data: model_full
## DW = 1.2157, p-value = 1.598e-06
## alternative hypothesis: true autocorrelation is greater than 0

sum(residuals(model_full))

## [1] -1.465494e-14

bptest(model_full)

##
## studentized Breusch-Pagan test
##
## data: model_full
## BP = 22.818, df = 23, p-value = 0.4714

white_test(model_full)

## White's test results
##
## Null hypothesis: Homoskedasticity of the residuals
## Alternative hypothesis: Heteroskedasticity of the residuals
## Test Statistic: 4.24
## P-value: 0.119811

y <- df$VISA_Price
yh <- fitted(model_full)
```

```

SST_full <- sum((y - mean(y))^2)
SSR_full <- deviance(model_full)
SSE_full <- SST_full - SSR_full
SSR_full

## [1] 8147.516

# 6. Reduced Model
model_visa_reduced <- lm(
  VISA_Price ~
    SP500_Return + Nasdaq_Return + VIX_Level +
    Oil_Price +
    PCE_Consumption + Unemp_Rate +
    Y2Y +
    Financial_Conditions_Index +
    Payroll_Employment +
    Real_Disposable_Income +
    Consumer_Loans,
  data = df
)

summary(model_visa_reduced)

##
## Call:
## lm(formula = VISA_Price ~ SP500_Return + Nasdaq_Return + VIX_Level +
##      Oil_Price + PCE_Consumption + Unemp_Rate + Y2Y + Financial_Conditio
##      ns_Index +
##      Payroll_Employment + Real_Disposable_Income + Consumer_Loans,
##      data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -41.026  -5.937   2.843   8.810  31.227
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.035e+03  3.447e+02  -3.001  0.00406 **
## SP500_Return    2.777e+02  1.263e+02   2.198  0.03225 *
## Nasdaq_Return  -1.533e+02  1.009e+02  -1.519  0.13454
## VIX_Level       6.868e-01  5.657e-01   1.214  0.23002
## Oil_Price      -4.337e-01  2.289e-01  -1.894  0.06354 .
## PCE_Consumption 2.457e-02  5.423e-03   4.530 3.30e-05 ***
## Unemp_Rate      2.019e+01  3.503e+00   5.762 4.09e-07 ***
## Y2Y            -7.914e+00  4.406e+00  -1.796  0.07808 .
## Financial_Conditions_Index -4.398e+01  2.264e+01  -1.943  0.05727 .
## Payroll_Employment 6.599e-03  2.256e-03   2.925  0.00502 **
## Real_Disposable_Income -3.369e-03  3.749e-03  -0.899  0.37289
## Consumer_Loans  -7.039e-02  3.680e-02  -1.912  0.06113 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.92 on 54 degrees of freedom

```

```
## Multiple R-squared:  0.926, Adjusted R-squared:  0.911
## F-statistic: 61.47 on 11 and 54 DF,  p-value: < 2.2e-16
```

```
SSR_red <- deviance(model_visa_reduced)
SSR_red
```

```
## [1] 12026.7
```

```
# 7. Final Model
```

```
model_visa_final <- lm(
  VISA_Price ~
    PCE_Consumption +
    Unemp_Rate +
    Y2Y +
    Financial_Conditions_Index +
    Payroll_Employment,
  data = df
)
```

```
coef(model_visa_final)
```

```
##              (Intercept)              PCE_Consumption
##          -1.552471e+03              1.855736e-02
##              Unemp_Rate              Y2Y
##          2.103644e+01          -1.521980e+01
## Financial_Conditions_Index Payroll_Employment
##          -5.191875e+01              9.068307e-03
```

```
residuals(model_visa_final)
```

```
##          1          2          3          4          5          6
##  0.8073882 -5.2255033  7.8158049 14.3610808 -1.8794322 -19.9541425
##          7          8          9         10         11         12
## -30.4783382 10.1845900 -0.4986504 -7.0357647 20.1734586 25.3130371
##         13         14         15         16         17         18
## -1.9959752 18.0849425 -2.2042341  9.9589723  4.7782686 -0.2476465
##         19         20         21         22         23         24
## 13.7923327 -3.3642232 -5.8683434 -19.3901149 -34.3297652 -8.7985186
##         25         26         27         28         29         30
##  1.9197520 -3.8923197 11.1691733  5.6347149  4.8108392 -7.1315073
##         31         32         33         34         35         36
##  0.2734832 -11.1817878 -15.2155465  8.9805559 11.5130522  1.6040499
##         37         38         39         40         41         42
##  5.8900278  1.6236176  2.5127130  5.1710120 -8.1481000 10.0383245
##         43         44         45         46         47         48
##  6.4101665  4.4642587 -12.4346368 -10.3141185  6.0923916 -6.7639960
##         49         50         51         52         53         54
##  5.6535017 10.6608046  1.3112409 -3.9633763 -7.4067789 -23.1196208
##         55         56         57         58         59         60
## -30.2110764 -27.9429872 -37.7579345 -18.0455756 -0.5628461 -2.8315257
##         61         62         63         64         65         66
## 23.0924635 38.3852133 22.0613909 11.0924960 31.5222408 11.0370268
```

```
summary(model_visa_final)
```

```
##
## Call:
## lm(formula = VISA_Price ~ PCE_Consumption + Unemp_Rate + Y2Y +
##      Financial_Conditions_Index + Payroll_Employment, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.758  -7.108   1.059   9.714  38.385
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.552e+03  1.924e+02  -8.068 3.75e-11 ***
## PCE_Consumption    1.856e-02  3.852e-03   4.818 1.03e-05 ***
## Unemp_Rate       2.104e+01  2.196e+00   9.579 1.06e-13 ***
## Y2Y             -1.522e+01  2.938e+00  -5.181 2.72e-06 ***
## Financial_Conditions_Index -5.192e+01  1.562e+01  -3.323  0.00152 **
## Payroll_Employment    9.068e-03  1.515e-03   5.986 1.30e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.57 on 60 degrees of freedom
## Multiple R-squared:  0.9105, Adjusted R-squared:  0.903
## F-statistic: 122.1 on 5 and 60 DF, p-value: < 2.2e-16

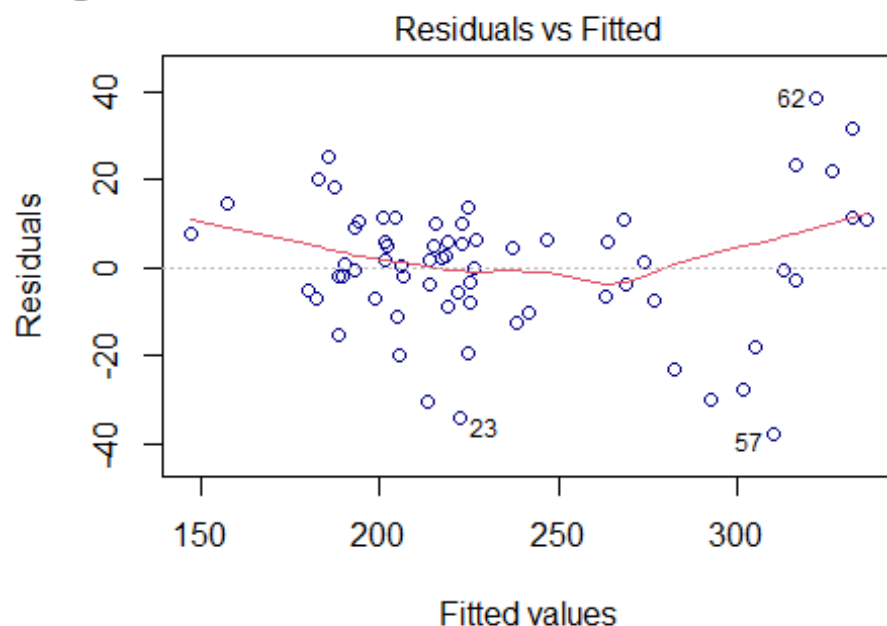
SSR_final <- deviance(model_visa_final)
SSR_final

## [1] 14554.64

y <- df$VISA_Price
yhf <- fitted(model_visa_final)

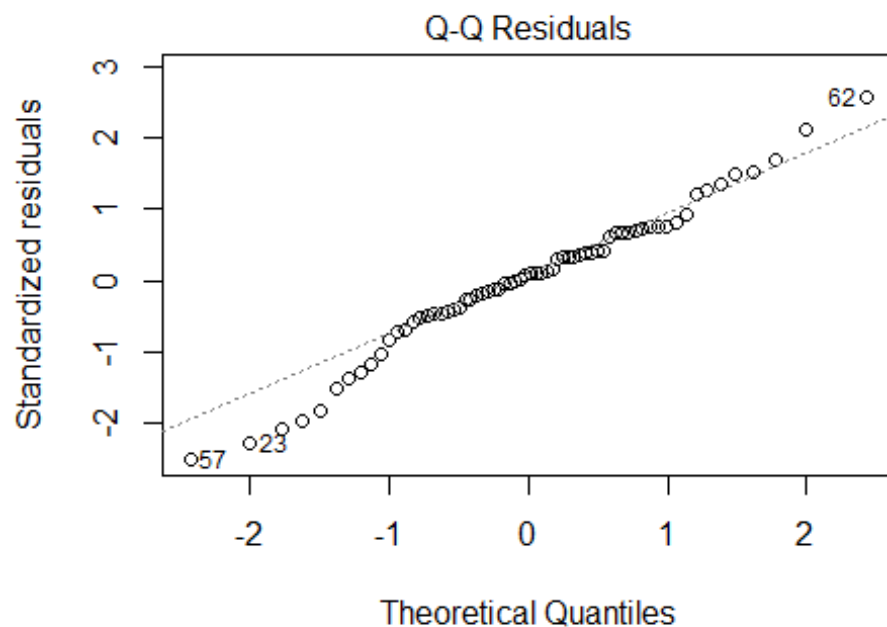
# 8. Checking Gauss Markov assumptions again on the Final Model
plot(model_visa_final, which = 1, col = "darkblue", sub = "",
      main = "Figure C1: Residuals vs Fitted Values for Final Model")
```

Figure C1: Residuals vs Fitted Values for Final Model



```
plot(model_visa_final, which = 2, col = "black", sub = "",
      main = "Figure C2: QQ Plot of Standardized Residuals for Final Model"
)
```

Figure C2: QQ Plot of Standardized Residuals for Final Model



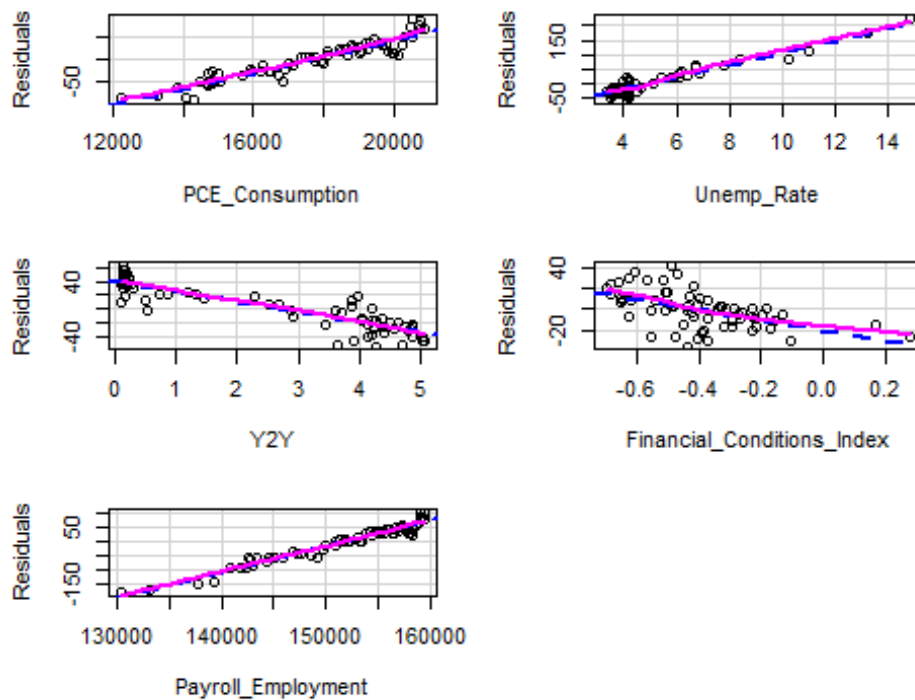
```
vif(model_visa_final)
```

##	PCE_Consumption	Unemp_Rate
##	19.320279	6.548957

```
##          Y2Y Financial_Conditions_Index
##          8.741402          2.367230
##      Payroll_Employment
##          29.291030

car::crPlots(model_visa_final, ylab = "Residuals",
             main = "Figure C3: Component plus Residual Plots for Final Mo
del")
```

Figure C3: Component plus Residual Plots for Final Mode



```
dwtest(model_visa_final)

##
## Durbin-Watson test
##
## data: model_visa_final
## DW = 0.90101, p-value = 2.056e-08
## alternative hypothesis: true autocorrelation is greater than 0

sum(residuals(model_visa_final))

## [1] 3.300138e-14

bptest(model_visa_final)

##
## studentized Breusch-Pagan test
##
## data: model_visa_final
## BP = 16.644, df = 5, p-value = 0.005228

white_test(model_visa_final)
```

```

## White's test results
##
## Null hypothesis: Homoskedasticity of the residuals
## Alternative hypothesis: Heteroskedasticity of the residuals
## Test Statistic: 13.01
## P-value: 0.001497

# 9. Model Comparison Table
mtable(
  "Full Model"      = model_full,
  "Reduced Model"   = model_visa_reduced,
  "Final Model"     = model_visa_final
)

##
## Calls:
## Full Model: lm(formula = VISA_Price ~ VISA_Volume + SP500_Return + Nasdaq_Return +
##      VIX_Level + DXY_Return + Gold_Return + Bitcoin_Return + Oil_Price +
##      Retail_Sales + PCE_Consumption + Industrial_Production +
##      Unemp_Rate + Personal_Income + Consumer_Sentiment + Consumer_Loans
##      +
##      CPI_Index + CPI_Mom + Y10Y + Y2Y + Financial_Conditions_Index +
##      Payroll_Employment + Real_Disposable_Income + Housing_Starts,
##      data = df)
## Reduced Model: lm(formula = VISA_Price ~ SP500_Return + Nasdaq_Return +
##      VIX_Level +
##      Oil_Price + PCE_Consumption + Unemp_Rate + Y2Y + Financial_Conditions_Index +
##      Payroll_Employment + Real_Disposable_Income + Consumer_Loans,
##      data = df)
## Final Model: lm(formula = VISA_Price ~ PCE_Consumption + Unemp_Rate + Y2Y +
##      Financial_Conditions_Index + Payroll_Employment, data = df)
##
## =====
##
##                               Full Model   Reduced Model   Final Model
## -----
##
## (Intercept)                -2054.746*    -1034.528**    -1552.471***
##                               (834.683)    (344.691)    (192.420)
## VISA_Volume                  -0.000
##                               (0.000)
## SP500_Return                 35.089        277.720*
##                               (149.183)    (126.347)
## Nasdaq_Return               -10.783       -153.329
##                               (113.781)    (100.925)
## VIX_Level                   -0.208         0.687
##                               (0.754)    (0.566)
## DXY_Return                  -198.634
##                               (143.232)
## Gold_Return                  -66.404
##                               (57.955)

```

```
## Bitcoin_Return      2.011
##                    (14.738)
## Oil_Price           -1.085**      -0.434
##                    (0.339)      (0.229)
## Retail_Sales         0.001*
##                    (0.000)
## PCE_Consumption     -0.103*      0.025***      0.019***
##                    (0.039)      (0.005)      (0.004)
## Industrial_Production 2.895
##                    (3.097)
## Unemp_Rate          23.196***      20.188***      21.036***
##                    (5.755)      (3.503)      (2.196)
## Personal_Income      0.050
##                    (0.025)
## Consumer_Sentiment   0.543
##                    (0.450)
## Consumer_Loans       -0.086      -0.070
##                    (0.044)      (0.037)
## CPI_Index            4.897
##                    (2.860)
## CPI_Mom              -4.726
##                    (10.132)
## Y10Y                 15.343
##                    (11.679)
## Y2Y                  -23.277*      -7.914      -15.220***
##                    (9.867)      (4.406)      (2.938)
## Financial_Conditions_Index -117.647**      -43.982      -51.919**
##                    (37.460)      (22.639)      (15.622)
## Payroll_Employment   0.012**      0.007**      0.009***
##                    (0.004)      (0.002)      (0.002)
## Real_Disposable_Income -0.057*      -0.003
##                    (0.027)      (0.004)
## Housing_Starts       -0.025
##                    (0.026)
```

```
## -----
--
## R-squared           0.950           0.926           0.911
## N                   66              66              66
```

```
## =====
==
```

Significance: *** = p < 0.001; ** = p < 0.01; * = p < 0.05

#####

B) TIME SERIES SECTION - VISA

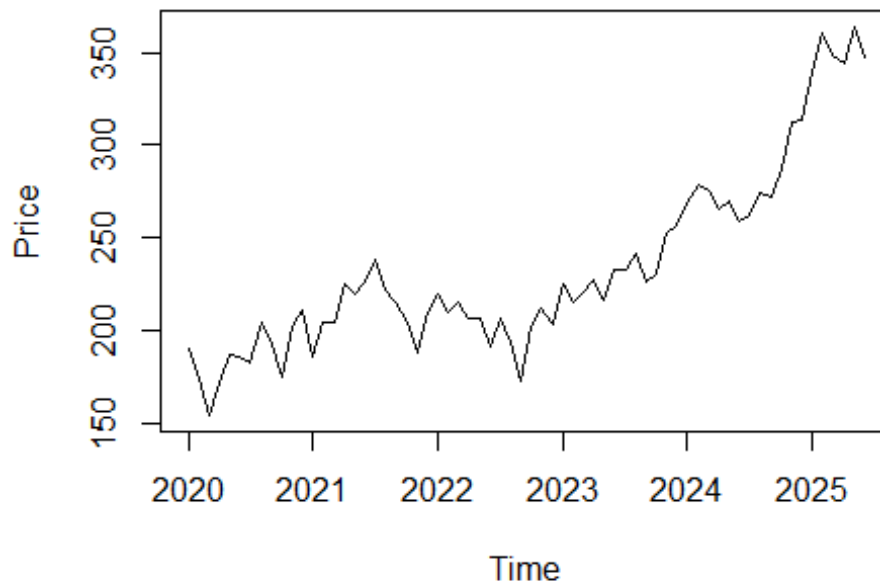
#####

1. Convert VISA monthly price to a time series

```
visa_prices_ts <- ts(df$VISA_Price, start = c(2020, 1), frequency = 12)
```

```
plot(visa_prices_ts, type = "l",
     main = "Figure D1: VISA Monthly Adjusted Price Series",
     xlab = "Time", ylab = "Price")
```

Figure D1: VISA Monthly Adjusted Price Series



```
adf.test(visa_prices_ts)

##
## Augmented Dickey-Fuller Test
##
## data:  visa_prices_ts
## Dickey-Fuller = -1.2593, Lag order = 4, p-value = 0.8756
## alternative hypothesis: stationary

kpss.test(visa_prices_ts)

## Warning in kpss.test(visa_prices_ts): p-value smaller than printed p-value
##
## KPSS Test for Level Stationarity
##
## data:  visa_prices_ts
## KPSS Level = 1.358, Truncation lag parameter = 3, p-value = 0.01

# 2. Convert to log returns and split into training/testing
visa_logret_ts <- diff(log(visa_prices_ts))
visa_logret_ts <- na.omit(visa_logret_ts)

n_train <- 52
train_ts <- window(visa_logret_ts, end = time(visa_logret_ts)[n_train])
test_ts <- window(visa_logret_ts, start = time(visa_logret_ts)[n_train + 1])

length(train_ts)

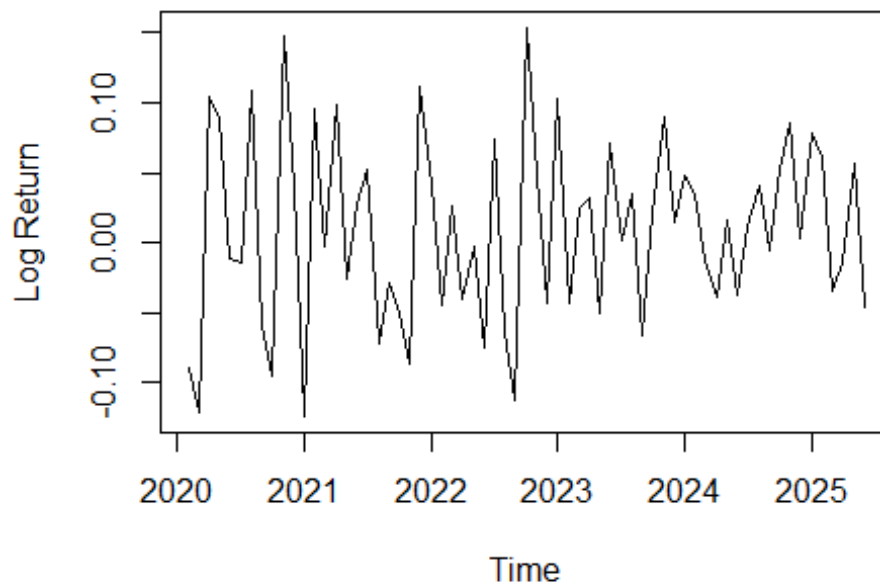
## [1] 52
```

```
length(test_ts)

## [1] 13

# 3. Stationarity of Training Set
plot(visa_logret_ts, type = "l",
     main = "Figure D2: VISA Monthly Log Returns",
     xlab = "Time", ylab = "Log Return")
```

Figure D2: VISA Monthly Log Returns



```
adf.test(visa_logret_ts)

## Warning in adf.test(visa_logret_ts): p-value smaller than printed p-value
##
## Augmented Dickey-Fuller Test
##
## data:  visa_logret_ts
## Dickey-Fuller = -4.2315, Lag order = 3, p-value = 0.01
## alternative hypothesis: stationary

kpss.test(visa_logret_ts)

## Warning in kpss.test(visa_logret_ts): p-value greater than printed p-value
##
## KPSS Test for Level Stationarity
##
## data:  visa_logret_ts
## KPSS Level = 0.13948, Truncation lag parameter = 3, p-value = 0.1
```

```

# 4. AR(1)
ar1_visa <- arima(train_ts, order = c(1, 0, 0))
ar1_visa

##
## Call:
## arima(x = train_ts, order = c(1, 0, 0))
##
## Coefficients:
##          ar1  intercept
##       -0.2313    0.0070
## s.e.   0.1360    0.0077
##
## sigma^2 estimated as 0.004681:  log likelihood = 65.66,  aic = -125.32

# 5. ARMA(1,1)
arma11_visa <- arima(train_ts, order = c(1, 0, 1))
arma11_visa

##
## Call:
## arima(x = train_ts, order = c(1, 0, 1))
##
## Coefficients:
##          ar1         ma1  intercept
##       0.3110  -0.7100    0.008
## s.e.  0.2658   0.1938    0.004
##
## sigma^2 estimated as 0.004216:  log likelihood = 68.25,  aic = -128.5

# 6. ARIMA(1,1,1)
arima111_visa <- arima(train_ts, order = c(1, 1, 1))
arima111_visa

##
## Call:
## arima(x = train_ts, order = c(1, 1, 1))
##
## Coefficients:
##          ar1         ma1
##       -0.2160  -1.0000
## s.e.   0.1382   0.0577
##
## sigma^2 estimated as 0.004775:  log likelihood = 61.73,  aic = -117.45

# 7. ARMA(1,1) + GARCH(1,1)
garch_visa <- garchFit(~ arma(1, 1) + garch(1, 1), data = train_ts,
                      cond.dist = "norm")

##
## Series Initialization:
##  ARMA Model:          arma
##  Formula Mean:        ~ arma(1, 1)
##  GARCH Model:         garch
##  Formula Variance:    ~ garch(1, 1)

```

```

## ARMA Order:          1 1
## Max ARMA Order:      1
## GARCH Order:         1 1
## Max GARCH Order:     1
## Maximum Order:       1
## Conditional Dist:     norm
## h.start:             2
## llh.start:           1
## Length of Series:    52
## Recursion Init:      mci
## Series Scale:        0.07101322
##
## Parameter Initialization:
## Initial Parameters:   $params
## Limits of Transformations: $U, $V
## Which Parameters are Fixed? $includes
## Parameter Matrix:
##           U           V      params includes
## mu      -0.93632600   0.936326  0.1120971    TRUE
## ar1     -0.99999999   1.000000  0.3110436    TRUE
## ma1     -0.99999999   1.000000 -0.7099813    TRUE
## omega   0.00000100  100.000000  0.1000000    TRUE
## alpha1  0.00000001   1.000000  0.1000000    TRUE
## gamma1 -0.99999999   1.000000  0.1000000    FALSE
## beta1   0.00000001   1.000000  0.8000000    TRUE
## delta   0.00000000   2.000000  2.0000000    FALSE
## skew    0.10000000  10.000000  1.0000000    FALSE
## shape   1.00000000  10.000000  4.0000000    FALSE
## Index List of Parameters to be Optimized:
## mu      ar1      ma1  omega alpha1 beta1
## 1        2        3    4      5      7
## Persistence:          0.9
##
## --- START OF TRACE ---
## Selected Algorithm: nlminb
##
## R coded nlminb Solver:
##
## 0:      68.157513: 0.112097 0.311044 -0.709981 0.100000 0.100000 0.800
000
## 1:      67.909468: 0.112055 0.310688 -0.707457 0.0916258 0.0925157 0.7
91858
## 2:      67.861120: 0.111868 0.309422 -0.696142 0.0885732 0.0857592 0.7
88608
## 3:      67.801624: 0.111655 0.306756 -0.685460 0.0962974 0.0866358 0.7
92116
## 4:      67.749961: 0.111356 0.299193 -0.676994 0.0964812 0.0795276 0.7
89040
## 5:      67.712815: 0.111043 0.289506 -0.671475 0.102910 0.0759015 0.79
2076
## 6:      67.676776: 0.110766 0.280665 -0.668065 0.104921 0.0660801 0.79
1752
## 7:      67.644516: 0.110540 0.275225 -0.664277 0.112737 0.0576931 0.79

```

```

5981
## 8: 67.616809: 0.110310 0.269238 -0.661858 0.115735 0.0459177 0.79
7931
## 9: 67.591936: 0.110013 0.258750 -0.661428 0.120412 0.0400794 0.80
2718
## 10: 67.518034: 0.108662 0.240384 -0.639578 0.127428 1.00000e-08 0.
828157
## 11: 67.472769: 0.107855 0.213628 -0.637040 0.125612 1.00000e-08 0.
835690
## 12: 67.472733: 0.107793 0.212553 -0.635476 0.124418 1.00000e-08 0.
835235
## 13: 67.471352: 0.107779 0.212338 -0.635268 0.125177 1.00000e-08 0.
836089
## 14: 67.470866: 0.107743 0.211669 -0.634501 0.124697 1.00000e-08 0.
836017
## 15: 67.470367: 0.107701 0.211157 -0.634227 0.124600 1.00000e-08 0.
836952
## 16: 67.469950: 0.107657 0.210610 -0.633840 0.123781 1.00000e-08 0.
837237
## 17: 67.469508: 0.107614 0.210172 -0.633560 0.123593 1.00000e-08 0.
838183
## 18: 67.469120: 0.107569 0.209725 -0.633177 0.122751 1.00000e-08 0.
838559
## 19: 67.468717: 0.107526 0.209373 -0.632891 0.122498 1.00000e-08 0.
839524
## 20: 67.468355: 0.107482 0.208990 -0.632523 0.121648 1.00000e-08 0.
839965
## 21: 67.467980: 0.107439 0.208685 -0.632248 0.121345 1.00000e-08 0.
840940
## 22: 67.467640: 0.107396 0.208342 -0.631904 0.120492 1.00000e-08 0.
841434
## 23: 67.467288: 0.107354 0.208066 -0.631645 0.120147 1.00000e-08 0.
842414
## 24: 67.466965: 0.107313 0.207750 -0.631325 0.119294 1.00000e-08 0.
842953
## 25: 67.466631: 0.107273 0.207496 -0.631081 0.118912 1.00000e-08 0.
843936
## 26: 67.466322: 0.107233 0.207202 -0.630781 0.118061 1.00000e-08 0.
844513
## 27: 67.392592: 0.103272 0.193047 -0.621487 0.0399305 1.00000e-08 0
.942195
## 28: 67.349525: 0.102556 0.190899 -0.618614 0.0227141 1.00000e-08 0
.967358
## 29: 67.208464: 0.101832 0.188584 -0.615805 0.00363025 1.00000e-08
0.991094
## 30: 67.144600: 0.101264 0.196226 -0.610801 1.00000e-06 1.00000e-08
0.994649
## 31: 67.143626: 0.102618 0.189166 -0.612929 1.00000e-06 1.00000e-08
0.993973
## 32: 67.139342: 0.104064 0.192812 -0.609472 1.00000e-06 1.00000e-08
0.993920
## 33: 67.136253: 0.103234 0.207217 -0.618506 1.00000e-06 1.00000e-08
0.994183
## 34: 67.135416: 0.105833 0.202540 -0.616783 1.00000e-06 1.00000e-08

```

```

0.994265
## 35:      67.135150: 0.105405 0.204397 -0.619056 1.00000e-06 1.00000e-08
0.994146
## 36:      67.135146: 0.105336 0.204458 -0.619307 1.00000e-06 1.00000e-08
0.994160
## 37:      67.135146: 0.105345 0.204454 -0.619303 1.00000e-06 1.00000e-08
0.994160
## 38:      67.135146: 0.105345 0.204454 -0.619303 1.00000e-06 1.00000e-08
0.994160
##
## Final Estimate of the Negative LLH:
## LLH:  -70.3991      norm LLH:  -1.353829
##          mu          ar1          ma1          omega          alpha1
## 7.480859e-03 2.044539e-01 -6.193029e-01 5.042877e-09 1.000000e-08
##          beta1
## 9.941600e-01
##
## R-optimhess Difference Approximated Hessian Matrix:
##          mu          ar1          ma1          omega          alp
ha1
## mu      -87885.94690    -711.71232      82.18684 -4.049139e+05 -2.432778e
+03
## ar1      -711.71232    -65.82625     -53.67072 -3.128336e+04 -7.945175e
+01
## ma1       82.18684     -53.67072     -86.47583 -3.780428e+04 -8.475278e
+01
## omega  -404913.91428 -31283.36477 -37804.27836 -1.452559e+09 -6.588880e
+06
## alpha1  -2432.77803     -79.45175     -84.75278 -6.588880e+06 -2.998177e
+04
## beta1   -1041.95733    -104.41064    -132.97625 -5.617129e+06 -2.554827e
+04
##          beta1
## mu      -1041.9573
## ar1      -104.4106
## ma1       -132.9762
## omega  -5617128.8007
## alpha1  -25548.2709
## beta1   -21592.9406
## attr(,"time")
## Time difference of 0.01049304 secs
##
## --- END OF TRACE ---
##
##
## Time to Estimate Parameters:
## Time difference of 0.06020212 secs

garch_visa

##
## Title:
## GARCH Modelling
##

```

```
## Call:
## garchFit(formula = ~arma(1, 1) + garch(1, 1), data = train_ts,
## cond.dist = "norm")
##
## Mean and Variance Equation:
## data ~ arma(1, 1) + garch(1, 1)
## <environment: 0x0000016262309dc8>
## [data = train_ts]
##
## Conditional Distribution:
## norm
##
## Coefficient(s):
##          mu          ar1          ma1          omega          alpha1
beta1
## 7.4809e-03 2.0445e-01 -6.1930e-01 5.0429e-09 1.0000e-08 9.941
6e-01
##
## Std. Errors:
## based on Hessian
##
## Error Analysis:
##      Estimate Std. Error t value Pr(>|t|)
## mu      7.481e-03 4.165e-03 1.796 0.07247 .
## ar1     2.045e-01 2.333e-01 0.877 0.38075
## ma1     -6.193e-01 2.285e-01 -2.711 0.00671 **
## omega   5.043e-09 1.329e-03 0.000 1.00000
## alpha1  1.000e-08 2.215e-01 0.000 1.00000
## beta1   9.942e-01 3.461e-02 28.726 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log Likelihood:
## 70.3991 normalized: 1.353829
##
## Description:
## Sat Nov 29 19:18:06 2025 by user: Deepak

# 8. Forecasts
ar1_pred      <- predict(ar1_visa, n.ahead = length(test_ts))$pred
arma11_pred   <- predict(arma11_visa, n.ahead = length(test_ts))$pred
arima111_pred <- predict(arima111_visa, n.ahead = length(test_ts))$pred
garch_pred    <- predict(garch_visa, n.ahead = length(test_ts))$meanF
orecast
actual        <- as.numeric(test_ts)

# 9. Forecast-based SSR
SSR_ar1_forecast <- sum((ar1_pred - actual)^2)
SSR_arma11_forecast <- sum((arma11_pred - actual)^2)
SSR_arima111_forecast <- sum((arima111_pred - actual)^2)
SSR_garch_forecast <- sum((garch_pred - actual)^2)

# 10. Final Comparison Table
SSR_forecast_comparison <- data.frame(
```

```

Model = c("AR(1)", "ARMA(1,1)", "ARIMA(1,1,1)", "ARMA(1,1) + GARCH(1,1)"
),
Forecast_SSR = c(
  SSR_ar1_forecast,
  SSR_arma11_forecast,
  SSR_arma111_forecast,
  SSR_garch_forecast
)
)

```

```

print(SSR_forecast_comparison)

```

##	Model	Forecast_SSR
## 1	AR(1)	0.02695904
## 2	ARMA(1,1)	0.02698364
## 3	ARIMA(1,1,1)	0.02697507
## 4	ARMA(1,1) + GARCH(1,1)	0.02702092